

REVIEW

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Progress in thermal energy storage technologies for achieving carbon neutrality

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Abstract

China is committed to the targets of achieving peak CO₂ emissions around 2030 and realizing carbon neutrality around 2060. To realize carbon neutrality, people are seeking to replace fossil fuel with renewable energy. Thermal energy storage is the key to overcoming the intermittence and fluctuation of renewable energy utilization. In this paper, the relation between renewable energy and thermal energy storage is first addressed. Then, the classifications of thermal energy storage and Carnot batteries are given. The aim of this review is to provide an insight into the promising thermal energy storage technologies for the application of renewable energy in order to realize carbon neutrality. Three types of heat storage methods, especially latent heat storage and thermochemical heat storage, are analyzed in detail. The application of thermal energy storage is influenced by many heat storage properties, such as temperature range, heat storage capacity, cost, stability, and technical readiness. Therefore, the heat storage properties for different heat storage technologies are reviewed and compared. The advantage and challenge of different heat storage technologies and Carnot batteries for carbon neutrality processes are analyzed. Finally, the prospects of different heat storage technologies are summarized.

Keywords Thermal energy storage, Carnot batteries, Carbon neutrality, Heat transfer, Cycling stability

1 Introduction

1.1 Energy consumption and CO₂ emission

Global warming is an important environmental issue. According to the sixth assessment report of the Intergovernmental Panel on Climate Change (IPCC), it is very likely that human influence has contributed to the observed global-scale changes in the frequency and intensity of daily temperature extremes since the mid-twentieth century [1]. The influence of carbon dioxide (CO₂) on global warming has been demonstrated. Therefore, many countries have joined the Paris Agreement and announce that they will realize carbon neutrality around the year 2050 or 2060.

China is committed to the targets of achieving peak CO₂ emissions around 2030 and realizing carbon neutrality around 2060 [2, 3]. However, the primary global energy consumption mainly depends on fossil fuels [4, 5]. The total amount of coal, oil, and gas consumption was about 136,000 TWh in 2021, which is much greater than that of renewable energy [5]. The use of huge amounts of fossil fuels results in a serious CO₂ emission problem. The total CO₂ emissions exceed 35 billion t [6]. According to the IEA data, the global CO₂ emissions have not yet peaked [7].

1.2 Renewable energy and energy storage

To realize carbon neutrality, people are trying to replace fossil fuels with renewable energy. There are many potential renewable energy options including wave, tidal, wind, solar thermal, biomass, photovoltaics, geothermal and hydropower [8]. Solar and wind power is widely used with its world's total installed capacity growing significantly [9]. However, renewable energy, such as

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solar and wind power, has inherent intermittency and fluctuation properties [10, 11]. To resolve these problems and improve the stability of renewable energy, energy storage is necessary [12].

The energy storage system can also promote efficient energy use in the field of daily life and industry, which is helpful for low carbon emissions [13]. At present, the main energy storage methods are electrochemical energy storage (lithium-ion batteries, lead-acid batteries, flow batteries, sodium-sulphur batteries, etc.) [14], thermal energy storage, fuel cells, hydrogen energy storage, flywheel storage, and pumped hydroelectric energy storage [13, 15]. Electrochemical energy storage is widely used and flexible. However, electrochemical energy storage has, especially lithium-ion batteries, high cost and safety problems [15, 16], which limits the large-scale application of this technology. Pumped hydroelectric energy storage is currently the most mature technology, which can satisfy large-scale energy storage, but geological and geographical conditions restrict its application [13]. The greatest number of operational projects is battery energy storage technology. The number of pumped hydroelectric energy storage projects is second and the thermal system follows [16].

Thermal energy storage is a good choice for large-scale and low-cost applications [12, 17]. For instance, Carnot batteries have the advantages in terms of simultaneous co-generation of thermal energy and power on the demand side as well as a very extensive temperature range of possible applications, making them an extremely promising option. The energy in thermal energy storage system can be used as heat directly [16]. At the same time, the final energy consumption is mostly used in form of thermal energy. According to the IEA data, heating for homes, industry and other applications accounts for around half of the total energy consumption [7]. Therefore, thermal energy storage can have a broad prospect in the future, and will have an important role in low carbon emissions.

1.3 Thermal energy storage

There are many thermal energy storage technologies that can be classified according to storage mechanism, temperature range, and others.

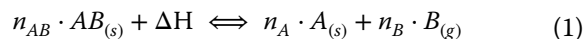
1.3.1 Classification according to energy storage mechanism

According to the different storage mechanisms, thermal energy storage can be divided into three types: sensible heat storage (SHS), latent heat storage (LHS), and thermochemical heat/energy storage (TCES) [13, 18]. The amount of sensible heat is related to the specific heat capacity of the materials and their temperature changes. Water, oil, and crushed stone are often used as SHS

materials [13]. The advantages of SHS are inexpensive, stable and convenient, but the big disadvantage is low energy density [13, 19].

Latent heat is stored between two changing phases of a material. The LHS materials are usually called phase change materials (PCMs) [13]. In the application, the heat stored in PCMs also includes solid sensible heat and liquid sensible heat besides latent heat [20]. Compared to the SHS, LHS has a good energy storage density [18]. Currently, LHS is gaining the most attention with increasing applications [13]. PCMs can be mainly divided into two groups: inorganic substances and organic substances [19].

Thermochemical energy storage (TCES) can convert thermal energy into chemical energy. Gas–solid TCES is often used because the reactants are easy to isolate. A typical TCES process can be described as [21]:



When the solid compound AB absorbs heat, it can be split into solid material A and gaseous material B. The heat can be stored at the ambient temperature for a long period due to thermal energy stored as chemical energy. Usually, the chemical reaction energy is larger than sensible heat and latent heat. TCES has the greatest energy density among the three thermal storage technologies, but there is a complicated control process in the application of heat storage [20, 22]. TCES can be divided into two types: chemical sorption and chemical reaction [23, 24]. Some researchers treat chemical sorption and chemical reaction as two different heat storage methods. However, chemical sorption also belongs to chemical reactions. We classify the two technologies as one method (TCES) in this paper. There are many chemical reaction materials used in heat storage, such as metal hydrides, metal hydroxides, metallic carbonates, metal oxides, etc. [22].

The comparison of energy density of different energy storage mechanisms is shown in Fig. 1 [23, 25]. Due to the energy density diversity, the volume of heat storage materials will significantly difference from each other. For example, to store 1850 kWh heat, the SHS (water) needs about 34 m³, while the TCES only needs about 1 m³.

1.3.2 Classification according to temperature range and other classifications

Considering the application (residential, industrial, and thermal power generation) and temperature characters of heat storage materials (evaporating point, melting point, decomposing temperature, etc.), thermal energy storage can also be classified according to the temperature range. The criteria of the temperature range are

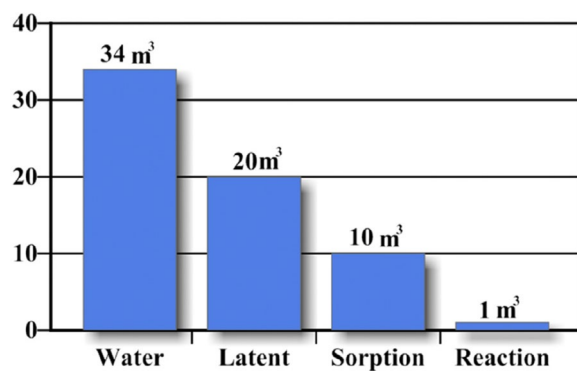


Fig. 1 Volume required to store 1850 kWh (With consideration of 25% heat losses, based on a 70 °C temperature increase for water) [23, 25]

non-uniform. For instance, some researchers divide the range into two main groups: low temperature (<200 °C) and high temperature (>200 °C) [17]. According to the heat source, three types (temperature ranges) are often used: below 150 °C, from 150 °C to 299 °C, and above 300 °C [20, 26]. There are also other classifications according to different temperatures [17, 19]. Considering the working temperature of heat storage and requirements of the temperature difference for heat transfer, the temperature of the heat storage materials in this paper are divided into three ranges: low-temperature ($T < 150$ °C), medium-temperature (150 °C $< T < 350$ °C), and high-temperature ($T > 350$ °C). In residential applications, domestic hot water is usually below 100 °C, and the vapor used in industry is normally lower than 300 °C [12]. If we are just considering the application, 100 °C and 300 °C are the suitable demarcation point. However, the temperature difference is required if we want the heat transfer from the heat storage materials to thermal consumer. Therefore, 50 °C temperature difference is added to the choosing temperatures.

From the view of application systems, the different heat storage systems can be classified into active and passive storage. The heat storage system circulates through a heat exchanger, and the system can realize both heat storage and release in the active system. By contrast, the passive system does not circulate through the system, and the heat transfer fluid is only for charging or discharging [17, 19].

Classifications according to material composition are also mainly used in many studies. However, this classification approach is usually listed in the subdirectory of different storage mechanism [19, 20].

1.4 Objectives and structure of the paper

As mentioned above, the global CO₂ emissions problem is serious, which leads to the huge requirement

for renewable energy. Energy storage is an indispensable part of the renewable energy process. Among those energy storage methods, thermal energy storage is inexpensive and can realize large-scale applications. Therefore, heat storage will play an important role in the future. This paper will discuss the thermal energy storage and their applications. As shown in Fig. 2, the structure of this paper has three main sections: SHS, LHS, and TCES. Every section includes some subdirectories. Finally, this review will summarize the findings and highlight the future research.

2 Sensible heat storage

SHS has become the most developed and widely used heat storage technology due to its simple principle and easy operation [27, 28]. The ideal SHS material should have good physical and chemical properties of large specific heat capacity, high density, high thermal conductivity, and low vapor pressure. Based on environmental and economic considerations, the materials should also be environmental friendly and low-cost [29]. However, almost no material can satisfy these requirements at the same time, so it is very important to select suitable heat storage materials in different situations.

2.1 Classification by temperature

According to the working temperature, SHS materials can be divided into three types: low-temperature ($T < 150$ °C), medium-temperature (150 °C $< T < 350$ °C) and high-temperature ($T > 350$ °C). The low-temperature materials are mainly represented by water. Medium-temperature materials mainly include various oils, such as mineral oil and synthetic thermal oil. Solid materials (concrete, rock, etc.) and various types of molten salt are the options of high-temperature SHS.

2.2 Classification by material

In practical application, different materials have obvious differences in system design, so SHS materials can be divided into liquid and solid materials.

2.2.1 Liquid material

Water, oil, and molten salt are often used as liquid SHS materials. Water is a clean and easily available material, and has a large specific heat capacity of 4.183 kJ/(kg·K). The boiling point of water is 100 °C, so it is not suitable for medium- and high-temperature heat storage fields. At present, water is considered as a potential material in seasonal heat storage due to its applicable physical and chemical properties [30]. Water is a simple material and has good cycling stability.

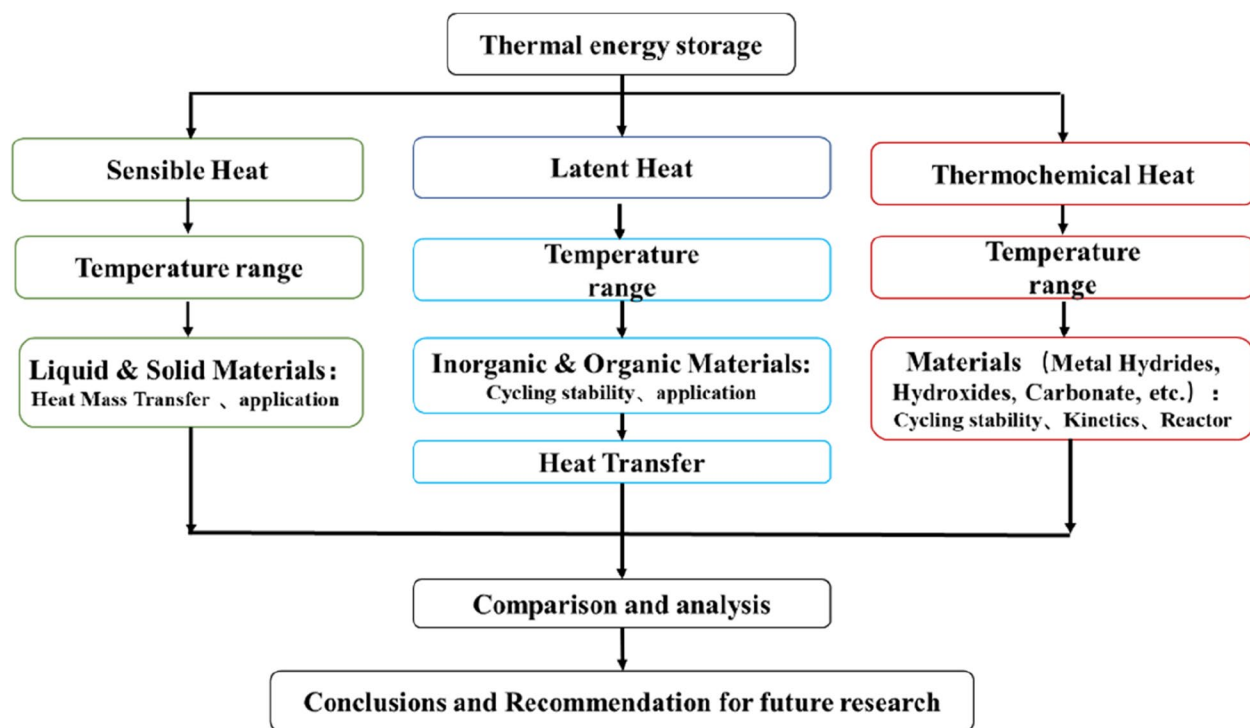


Fig. 2 Layout of the review paper

Liquid oil is also one of the SHS materials, which can be divided into mineral oil and synthetic heat conduction oil. The upper limit of the working temperature of mineral oil is 300 °C, which hinders the improvement of heat storage efficiency. The upper limit of heat storage temperature of thermal oil can reach about 400 °C, but the economic costs are relatively high. At the same time, high-temperature application scenarios will lead to toxic steam, inflammability, metamorphic, and other problems. Oil has good cycling stability and heat transfer performance below its decomposition temperature.

Molten salt has the advantages of high specific heat, density and heat capacity, and is widely used in thermochemical catalysts, green fuels, electrochemistry, heat carriers and other fields [31–33]. Pure molten salt is usually not suitable to use alone due to its fixed melting point, so the current molten salt heat storage research is mainly based on mixed molten salt such as solar salt ($\text{NaNO}_3\text{-KNO}_3$ in the proportions 60–40 wt.%) and Hitec salt ($\text{NaNO}_3\text{-KNO}_3\text{-NaNO}_2$ in the proportions 7-53-40 wt.%) [34–36].

Generally, the cycling stability and heat transfer performance of molten salt are more complicated. Ji et al. [37] encapsulated solar salt with SiO_2 nanoparticles and added graphene oxide to it. After 500 heating and cooling cycles, the specific heat capacity and thermal conductivity of the sample remained almost unchanged. Chen et al.

[38] studied the forced convection heat transfer of a molten salt nanofluid in a circular tube. It was found that the convective heat transfer coefficient increased by 39.9% with the addition of SiO_2 nanoparticles. They also investigated the heat transfer performance of Hitec salt under different sunlight heating angles in solar thermal power [39]. Molten salts are usually obtained from the liquid PCM, so more details of this part will be described in the Sections 3.2 and 3.3.

2.2.2 Solid materials

Solids as SHS materials have the advantages of high specific heat and low cost without considering steam pressure. The leakage problems can be easily solved by means of filter. Solid materials are commonly used to pack-beds and exchange heat with fluids [40, 41]. Santos et al. [42] built a thermal energy storage system, using hematite as storage material. The value of the specific heat of the hematite is 618 J/kg·K, and the thermal conductivity is 0.3 W/m·K. Schlipf et al. [43] further studied the feasibility of sand and stone for heat storage by theoretical and experimental analyses. Lucio-Martin et al. [44] analyzed the evolution of thermal properties of concrete during thermal cycling at 600 °C and found that the concrete mix ratio and the aggregates are the two factors affecting the thermal conductivity of concrete at high temperatures.

The cycling stability of solid SHS materials is also an important research topic. Becattini et al. [45] investigated the performance of six types of rocks of Alpine origin for 40 thermal cycles between 100 °C and 600 °C. The specific heat capacity of rocks decreased during the thermal cycle, while the porosity increased. After 120 thermal cycles, the mechanical properties of rhyolite and quartzite sandstone remained stable and had good resistance to thermal shock of heating and cooling. On the contrary, limestone, marble, and granite cracked due to mass loss or thermal expansion after thermal cycling [46]. The dolerite and hornfels were considered as potential materials for SHS after a 950 cycle test [47]. It is obvious that the cycling stability of solid SHS materials is quite different.

2.2.3 Material comparison

The comparison of different SHS materials is shown in Table 1, and the data is obtained around the year 2014–2017 according to the references. Water has a large specific heat, and it can be used below 100 °C. Oil can be used below 350 °C, and the specific heat of oil is lower than that of water, but it is higher than that of other materials. Rocks, sand, and most molten salt materials can be used above 500 °C. In general, SHS materials are widely used in the current industry and household application because of their stable properties and simple system.

2.3 Summary

SHS usually needs a large space to store the heat due to the low energy density. The great volume of the system will lead to bigger heat loss and a larger heat transfer area. Furthermore, the corrosion of molten salt is another

problem. New materials and novel heat transfer methods are the research highlights for SHS in the future.

3 Latent heat storage energy

LHS based on PCMs can offer high energy density and is considered to be a very attractive energy storage option. PCMs with solid–liquid phase changes are more efficient than liquid–vapor and solid–solid transitions [50].

Ideal PCMs should meet the following criteria: suitable melting temperature in the desired operating temperature range, large latent heat, high thermal conductivity, large specific heat capacity, good cycling and thermal stability, low corrosion, no phase separation and supercooling, no fire hazard and no toxicity, and low cost. However, it is difficult for each kind of PCM to satisfy all the criteria due to its unique thermophysical and chemical properties.

3.1 PCM classification based on melting temperature

As mentioned above, PCMs can be divided into low-temperature ($T_m < 150$ °C), medium-temperature (150 °C $\leq T_m \leq 350$ °C) and high-temperature ($T_m > 350$ °C) according to their melting temperatures. Low-temperature PCMs mainly include fatty acids (acetic acid, eladic acid, palmitic acid and so on), paraffin wax, hydrated salts represented by $MgCl_2 \cdot 6H_2O$, $CaCl_2 \cdot 12H_2O$, and $Na_2SO_4 \cdot 10H_2O$ as well as some metals and their alloys (Gallium and Bi-Pb eutectic). Detailed information about low-temperature PCMs (especially those PCMs with melting temperatures below 20 °C) for cold thermal energy storage was summarized by Oró et al. [51]. Medium-temperature PCMs mainly consist of some nitrates and their eutectics ($NaNO_3$ - $Ca(NO_3)_2$ eutectics, KNO_3 - $NaNO_2$ - $NaNO_3$), and so on. High-temperature PCMs are inorganic salts as well as their eutectics (K_2CO_3 , KF , LiF , $NaCl$, $MgCl_2$, Na_2CO_3 , $NaCl$ - $CaCl_2$ - $MgCl_2$ eutectics, etc.), metals (Al, Mg, etc.) and alloys (62.5% Al - 37.5% Mg, 82.5% Al - 17.5% Cu, 87% Al - 13% Si, etc.). The melting temperature range of PCMs is very wide, so they are applicable for various situations. The low-temperature PCMs are mainly used in the organic Rankine cycle with waste heat recovery and thermal energy storage systems for building heating and cooling application, while medium- or high-temperature PCMs can be used in solar power systems and other industrial applications [52].

3.2 PCMs classification based on material composition

According to the material composition, PCMs can be divided into two main categories: organic compounds and inorganic materials.

Table 1 Main characteristics of different SHS materials

Materials	Temperature		c_p (kJ/kg·K)	Average Cost (\$/kW·h)	Refs
	Low (°C)	High (°C)			
Water	0	100	4.183	-	[48]
Mineral oil	200	300	2.6	4.2	[19]
Synthetic oil	250	350	2.3	43	[19]
Rocks	-	-	0.96	-	[29]
Sand	-	-	0.83	4.2	[29]
Cast steel	200	700	0.60	60	[19]
Reinforced concrete	200	400	0.85	1.0	[19]
Solar salt	220	565	1.5	5.8	[49]
Hitec salt	142	535	1.4	10.7	[49]
Nitrite salts	250	450	1.5	12	[19]
Nitrate salts	265	565	1.6	3.7	[19]
Carbonate salts	450	850	1.8	11.0	[19]

3.2.1 Organic compounds

Material Organic materials can be further classified into two categories: paraffin and non-paraffin. They have the favorable characteristics of congruent melting and self-nucleation. Being non-corrosive and having good thermal stability also make organic PCMs an attractive option. However, flammability, possible toxicity, low thermal conductivity, and phase change enthalpy are some undesirable properties of this type of PCMs [53].

Paraffin wax consists of a mixture of mostly straight-chain n-alkanes ($\text{CH}_3-(\text{CH}_2)_n-\text{CH}_3$) and both the melting point and phase change enthalpy increase with the number of carbon atoms. Non-paraffin PCMs include fatty acids produced by $\text{CH}_3(\text{CH}_2)_{2n}\text{COOH}$, esters, and glycols [53]. Compared to paraffin wax, fatty acids have comparable latent heat but mild corrosive and high economic cost. Table 2 lists the detailed information of some organic PCMs (paraffins and non-paraffins), comparing their thermophysical properties. The phase change enthalpy of the organic materials is usually between 150 kJ/kg to 200 kJ/kg.

Application In current studies and applications, organic materials frequently play the role of building envelopes by themselves or mixed with concrete [57]. Zhang et al. [58] investigated PCMs incorporated in building envelopes for building's thermal management. However, most of the passive thermal energy storage systems are limited to short-term storage because they are uncontrollable and have low solar energy utilization efficiency. For seasonal storage situations, active storage combined with a solar collector system seems to have more potential. Hassan et al. [59] selected organic paraffin (RT-35HC) as PCM

to control the temperature of the photovoltaic panels, as shown in Fig. 3, and they found that its electrical efficiency was improved by 9.1% compared to conventional Photovoltaic. Öztürk [60] carried out a lab-scale seasonal LHS experiment (180 m² greenhouse located in Turkey) for greenhouse heating. This system has solar air collectors, a LHS unit containing 6000 kg of paraffin wax as the PCM, a lab-scale experimental greenhouse, a heat transfer component, and a data acquisition system. The results showed that this system can reach an acceptable energy efficiency (40.4%) and an average exergy efficiency (4.2%). Organic PCMs can also be used for the thermal management of photovoltaic panels to improve power generation efficiency. Qi et al. [61] simulated the performance of a space heating system with seasonal LHS and considered that it is a very efficient and promising building energy-saving technology. In a word, when used as PCMs,



Fig. 3 Experimental layout for thermal management of photovoltaic panels using paraffins [59]

Table 2 Thermophysical and chemical properties of organic PCMs [54–56]

Type	Material	P kg/m ³	T _m °C	Δh kJ/kg	K W/mK	c _p kJ/kgK	Corrosive	Toxicity
Paraffin wax	n-Tridecane	755	-5	160	-	-	-	Slight
	n-Hexadecane	773	18	237	-	-	-	slight
	n-Heneicosane	787	40.2	200	-	-	-	Non-toxic
	n-Tetracosane	799	50.6	255	-	-	-	NA
	n-Hexacosane	778	56.3	256	-	-	-	NA
	n-Octacosane	804	61.6	253	-	-	-	Slight
Non-paraffins	Acetic acid	1002	16.7	184	0.16	2.04 (l)	Mild	Non-toxic
	Lauric acid	1007	44	218	0.22	2.15 (l)	Mild	Acute
	Palmitic acid	989	61	222	0.21	2.20 (l)	Moderate	Non-toxic
	Thiosinamine	1167	77	140	-	-	Non-corrosive	High
	Catechol	1149	104.3	207	-	-	Strong	Slight
	Benzamide	1341	127.2	169.4	-	-	Non-corrosive	Acute

organic materials can usually be applied to building heating and cooling with construction in a passive way and seasonal storage purposes in an active way. However, the flammability, possible toxicity, low thermal conductivity, and low phase change enthalpy are restricting the application of organic materials.

3.2.2 Inorganic materials

Material Inorganic materials are a suitable medium/high-temperature PCM with wide sources, a high phase change enthalpy (68–1041 kJ/kg), a low vapor pressure and relatively low prices. However, inorganic salts and their eutectics usually suffer from poor thermal conductivity, corrosion problems, and subcooling/phase separation, which severely affect the efficiency and stability of LHS. Generally, inorganic materials can be divided into hydrated salts, metals, and alloys as well as other inorganic salts.

Hydrated salts are generally regarded as a typical crystalline solid composed of inorganic salt and water of which the chemical formula is $AB \cdot nH_2O$, where A and B represent the components of the salt and n represents the number of water molecules in the salt hydrate. The melting process of salt hydrates is essentially a process of dehydration to inorganic salt and water or low hydrate and water.

Most salt hydrates can be applied as low-temperature PCMs (< 150 °C) with a high phase change enthalpy from 125.9 kJ/kg to 301 kJ/kg. The thermophysical properties of several typical salt hydrates are listed in Table 3. Although salt hydrates exhibit better thermophysical properties than to organic PCMs, their shortcoming is also obvious: incongruent melting. Water extracted

from the melting process of salt hydrates does not fully dissolve the inorganic salts generated during the phase change process so that inorganic salts deposit at the bottom of the container, which makes a fully reversible solidification impossible due to the density difference. This problem is also known as phase separation.

In addition to the above phase separation problem, another main problem of salt hydrate is supercooling. Mechanical stirring, adding thickening agents, encapsulating PCMs [67], and modifying the chemical composition to make materials congruent [68] are effective methods to solve these problems. Kazemi and Mortazafi [69] used sodium tetra borate as a nucleating agent and successfully minimized the supercooling of Glauber's salt. Similarly, nitro aluminum was used as a nucleating agent to reduce supercooling [70], as were Si_3N_4 , ZrB_2 and SiO_2 , silver nanoparticles [71].

Metals and their alloys usually exhibit high heat storage density per unit volume, high thermal conductivity, sufficiently low degree of supercooling, better stability, smaller volume change, and relatively low vapor pressure [72]. Due to their relatively high costs and high corrosion for high-temperature application, less attention has been paid to metals and their alloys as PCMs. With the goal of zero carbon emissions, the development of renewable resources, the innovation of industrial waste heat recovery technology has diversified the demand for heat storage, which brought the unique value of metals and their alloys to attention. Thermophysical properties of selected melts and alloys are listed in Table 4.

Metals and metallic alloys as PCMs at high temperatures mainly focused on Al-based alloys in precious research. Birchenall et al. [74] investigated the thermal properties of several kinds of binary and ternary metal alloys containing elements Al, Si, Mg, Cu, and Zn. They found that

Table 3 Thermophysical properties of selected salt hydrates as PCMs

Material	ρ kg/m ³	T_m °C	Δh kJ/kg	k W/mK	c_p kJ/kgK	Refs
$Na_2SO_4 \cdot 10H_2O$	1485 (s)	31–32.4	251.1–254	0.544	1.93(s)	[62]
$Mn(NO_3)_2 \cdot 6H_2O$	1795 (s)	25.5–25.8	125.9–148	-	-	[63]
$CaCl_2 \cdot 6H_2O$	1682.4–1802 (s)	27.45–30	161.15–192	1.088(s)	1.4(s)	[56, 64]
$LiNO_3 \cdot 3H_2O$	1550 (s)	29.6–29.9	296	0.8 (s)	1.8 (s)	[65]
$Zn(NO_3)_2 \cdot 6H_2O$	1937–2065 (s)	36–36.4	134–147	0.464–0.469(l)	1.34(s)	[62]
$Fe(NO_3)_3 \cdot 9H_2O$	1684 (s)	47–47.2	155	-	-	[65]
$Na(CH_3COO) \cdot 3H_2O$	-	58–58.4	226–264	-	-	[54, 66]
$Ba(OH)_2 \cdot 8H_2O$	2070–2180 (s)	78	265.7–301	1.255(s)	1.17(s)	[65]
$Mg(NO_3)_2 \cdot 6H_2O$	1636–1640 (s)	89–95	254–267	0.611–0.669(s)	0.9(s)	[66]
$MgCl_2 \cdot 6H_2O$	1560–1570 (s)	115–117	195–172	0.694–0.704(s)	1.72–2.25(s)	[65]

Table 4 Thermophysical properties of selected metals and metal alloys as PCMs [66, 73]

Material	ρ kg/m ³	T_m °C	Δh kJ/kg	k W/mK	c_p kJ/kgK
Cesium	1796 (l)	28.65	16.40	17.4(l)	0.24(l)
Gallium	5903 (s)/6093 (l)	29.8–30	80.30	33.7 (s) 24 (l)	0.34 (s) 0.40 (l)
Rubidium	1470(l)	38.85	25.74	29.30 (l)	0.36 (l)
Potassium	664 (l)	63.2	59.59	54 (l)	0.78 (l)
Indium	7310	156.7	28.47	86	0.24
59% Al-35% Mg-6% Zn	2380	443	310	-	1.63
34.65% Mg-65.35% Al	2155	497	285	-	-
54% Al-22% Cu-18% Mg-6% Zn	3140	520	305	-	-
65% Al-30% Cu-5% Si	2730	571	422	-	-
88% Al-12% Si	2700	576	560	160	1.04
56% Cu-27% Si-17% Mg	4150	770	420	-	0.75

alloys with a large proportion of Al or Si had higher heat storage densities. In addition to aluminum-based alloys, magnesium-based alloys have also been studied. Fang et al. [75] compared the heat storage performance of three different kinds of Mg-Bi metallic alloys with different mass ratios, and they found that the latent heat and melting temperature ranged from 48.7 kJ/kg to 180.5 kJ/kg and from 546 °C to 548 °C, respectively.

Other inorganic salts mainly include pure carbonates, nitrates, chlorates and their eutectics. Table 5 lists the thermophysical properties of some pure inorganic salts as PCMs. However, due to the limited pure material types and wide temperature range of application, inorganic salt eutectics are needed to fill the uncovered temperature gaps.

Carbonates and their eutectics have a large heat capacity, a low viscosity, and are easily decomposed at high temperatures. They are attractive PCMs in the Concentrated

Solar Power (CSP) system with latent storage. Ren et al. [78] prepared 36 kinds of molten carbonate salts by mixing K_2CO_3 , Li_2CO_3 and Na_2CO_3 at different mass ratios, and they found that the melting points of most molten salts are around 400 °C. Six kinds of mixed salts were considered to be good options for PCMs in solar thermal power generation with considerably larger latent heat and lower costs (<\$100/kWh).

Nitrates and their eutectics usually melt at relatively low temperatures. These materials are easy to decompose above 500 °C with little corrosion and high thermal stability, which makes them ideal heat transfer fluids in CSP techniques [79]. Zhao et al. [80] prepared a $NaNO_3$ - $Ca(NO_3)_2$ eutectics mixture at the ratio of 3:7. The melting temperature and heat of fusion of this optimal proportion were 217.4 °C and 135.8 kJ/kg, respectively. More importantly, the cost was 35% lower than Solar Salt. The phase change characteristics, thermophysical properties of Solar Salt, Hitec and Na-K-Li-Ca nitrate as PCMs were

Table 5 Thermophysical properties of selected pure inorganic salts as PCMs

Material	ρ kg/m ³	T_m °C	Δh kJ/kg	K W/mK	c_p kJ/kgK	Refs
$AlCl_3$	2440	192	272–280	-	-	[56]
$LiNO_3$	2380	250–254	360–373	-	-	[56, 73]
$ZnCl_2$	2907	280	75	0.5 (s)	0.74 (l)	[76]
$NaNO_3$	2257–2261	306–310	172–199	0.5 (s)	1.1 (l)	[63]
KNO_3	2109–2110	330–336	88–266	0.5 (s)	0.935 (s)	[63]
$MgCl_2$	2140	714	452–454	-	-	[63, 73]
$NaCl$	2160	800–802	466.7–492	5 (s)	-	[63, 73]
LiF	-	848–868	932–1041	-	-	[56]
$NaCO_3$	2533	854–858	165–275.7	2 (s)	-	[63]
KF	2370	857–858	452–507	-	-	[63, 77]
K_2CO_3	2290	897–900	200–235.8	2 (s)	-	[73, 77]

studied by Iverson et al. [81]. Roget et al. [82] measured the melting temperature and enthalpy of KNO_3 - LiNO_3 eutectics and KNO_3 - NaNO_3 - LiNO_3 eutectics for LHS applications and found that the heat storage densities were 89 kWh/m^3 and 86 kWh/m^3 , respectively.

Compared to carbonates and nitrates, chlorides and their molten salts have severer corrosivity. However, chlorides have higher melting temperatures and abundant natural reserves such as NaCl , KCl and MgCl_2 . Chlorides and their eutectics are mainly researched as heat transfer fluids with an insufficient study on their characteristics as PCMs. Myers et al. [83] evaluated the melting temperatures, latent heat, and costs of 133 chloride salts and found that costs were lower than \$0.2/kJ latent heat. Du et al. [84] prepared NaCl - CaCl_2 - MgCl_2 eutectics with good thermal stability at high temperatures (500, 550, 600, 650 °C). Their melting temperatures and heat of fusion were 421 °C and 201.5 kJ/kg, respectively.

Cycling and thermal stability In melting and solidification cycles, some inorganic salts and their eutectics suffer from changes in microstructures as well as compositions. These salts may decompose in certain high-temperature applications, which could cause poor heat storage performance. This section concerns such cycling and thermal stability issues of these inorganic salts and composite PCMs.

Study on cycling stability of metals and alloys mainly focused on alloys, especially binary alloys and ternary alloys. Li et al. [85] investigated the cycle stability of Al-17 wt% Si metal alloys. They found that the latent heat and phase transition temperature of alloys increased due to the grain size and shape changes of Si after 1200 cycles. Zhao et al. [86] investigated the cycle stability of eutectic, hypoeutectic and hypereutectic Al-Si metal alloys and found that all of the melting temperatures remained stable after 100 circulations, but only eutectic Al-Si alloys and Al-20 wt% Si alloys had a slight variation in latent heat. Except for eutectic alloys, the thermal conductivity had variations after 100 cycles. For ternary alloys, Sun et al. [87] reported the cycling stability of Al-34 wt% Mg-6 wt% Zn eutectic metal alloys after 1000 cycles. Due to the degradation of the chemical structure, the latent heat decreased by 10.98%.

Research on the cycling and thermal stability of other inorganic salts mainly involved carbonates, nitrates, and chlorides. Carbonate salts and nitrate salts usually have relatively good cycling stability and lower decomposition temperatures. A small amount of additives can further help improve the stability of these inorganic salts. Table 6

summarizes the representative studies of the cycling and thermal stability of some carbonate salts, nitrate salts, and chloride salts.

Corrosion Jacob and Bruno [98] pointed out that steel-based metallic shells can be applied to encapsulate most high-temperature PCMs, which can tackle the corrosion problems, but the corrosion varies with the type of inorganic salts. The corrosion of salt hydrates is related to the type of hydrated salts as well as container materials. Cabeza et al. [99] compared the corrosion behaviour of EN AW-2007, E-Cu 57, Ms58 Flach, stainless steel and steel 37 K in different kinds of salt hydrates including $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ and $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ for 3–15 days. They reported that $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ was highly corrosive to metals and alloys, and the corresponding corrosion rate to aluminum as well as steel materials ranked high. Cabeza et al. [100] compared the corrosion of the same five metallic samples in sodium thiosulphate trihydrate as well as sodium acetate trihydrate and reported that stainless steel and aluminum exhibited better corrosion resistance than other metals.

With the increasing application temperature of metals and metal alloys, the corrosion of PCMs to containers and packaging materials should receive attention. Fukahori et al. [101] analyzed the corrosion of Al_2O_3 , AlN , Si_3N_4 , SiO_2 , and SiC in molten Al-Si alloys at the temperature of 1000 °C for 100 h. It was found that Al_2O_3 , and AlN as well as Si_3N_4 were highly compatible with Al-Si alloys, while SiO_2 eroded all samples. Zhao et al. [86] investigated the corrosion performance of eutectic, hypoeutectic, and hypereutectic Al-Si metals with six kinds of ceramics at 900 °C for 120 h.

Among the nitrate salts, Solar Salt was investigated in the field of corrosion. Good and Bradshaw [102] experimentally studied the effect of impurities in NaNO_3 - KNO_3 on the corrosion behaviour of ss304, ss316, and A36 carbon steel. They reported that the impurity contents had little effect on the corrosion behaviour and the corresponding corrosion rates were 6, 15, and 1–4 $\mu\text{m/yr}$, respectively, which were acceptable for engineering applications. Zhu et al. [103] evaluated the corrosion behaviour of ss316 in Hitec salts at 450 °C, 600 °C and 680 °C. Good corrosion resistance was observed at 450 °C. For chloride salts, Shankar et al. [104] investigated the corrosion of Ni-based alloys 600, 625, 690, and alloy 800H in molten LiCl - KCl eutectics at 400, 500 °C and 600 °C for 2 h. Compared to Ni-based alloy 625 and alloy 800H, Ni-based alloys 600 and 690 exhibited better corrosion resistance. Vignarooban et al. [105] estimated the corrosion behaviour of Hastelloys C-276, C-22, and

Table 6 Representative studies of the cycling and thermal stability of carbonates, nitrates and chlorides

Material	Heating conditions	Atmosphere	Cycles	Comments	Refs
LiKCO ₃	480–535 °C	-	129	Suitable for 8 kWh thermal energy storage application	[88]
Na ₂ CO ₃ - Li ₂ CO ₃	600 °C	CO ₂	500	Without weight loss, great thermal and chemical stability	[89]
32.1 wt% Li ₂ CO ₃ - 33.4 wt% Na ₂ CO ₃ - 34.5 wt% K ₂ CO ₃	Up to 1000 °C	N ₂	-	0.8% weight loss	[90]
		CO ₂		Stable	
		Ar		Decompose at 710–715 °C	
		Air		Decompose at 530 °C	
28.5 wt% Li ₂ CO ₃ - 71.5 wt% K ₂ CO ₃	300–600 °C	Air	40/100	A weakened sub-cooling	[91]
35 wt% Li ₂ CO ₃ - 65 wt% K ₂ CO ₃	~500 °C	Air	-	Unnoticeable phase change after a 20-cycle test, a weakened sub-cooling	[92]
22 wt% Li ₂ CO ₃ - 62 wt% K ₂ CO ₃ - 16 wt% Na ₂ CO ₃				Relatively stable over 350 °C, partial salt crept out in the liquid state, form a little nitrite (NaNO ₂ -NaNO ₃)	
NaNO ₃	~500 °C	Air	-	Unchanged latent heat, hardly changed chemical structures, less than 3% weight loss at 500 °C	[93]
Solar Salt	600 °C	Air	10–50	0.20% mass loss, high reproducibility of DSC curves in heating process	[80]
20 mol% Ca(NO ₃) ₂ - 80 mol% NaNO ₃	350–750 °C (thermal)	-	26	0.42% mass loss, better exothermic performance, half the cost of Solar Salt	[94]
30 mol% Ca(NO ₃) ₂ - 70 mol% NaNO ₃	80–350 °C (cycling)			Formed NO at 325 °C	
KNO ₃ - NaNO ₃ - LiNO ₃	30–1000 °C	Ar	-	Formed NO at 425 °C	[95]
		N ₂		Formed NO at 475 °C, limited long-term stability at 500 °C	
		Air		Formed NO at 540 °C	
		O ₂		31.3% reduction in latent heat	
68 wt% KNO ₃ - 18 wt% LiNO ₃ - 14 wt% Ca(NO ₃) ₂	400–500 °C, 32–72 days (thermal)	Air	800	Good thermal stability with 0–0.6 wt% CsNO ₃ at 400 °C but unstable at 500 °C, improved cycling stability with 0.2–0.38 wt% CsNO ₃ additives	[91]
Same ternary eutectics with 0–0.6 wt% CsNO ₃ additives	100–180 °C (cycling)			Salts crept out after 40 cycles	
64 wt% MgCl ₂ - 36 wt% KCl	300–600 °C	Air	40	No significant change	[96]
52 wt% MgCl ₂ - 48 wt% NaCl	600–650 °C, 2 h	Air	1000	No noticeable change in thermophysical properties	
Na ₂ CO ₃ - NaCl				Good thermal stability	[97]
Na ₂ SO ₄ - NaCl	550–680 °C, ≤ 6 h	Air	100	Reduced undercooling	
Same sample with α-alumina or mullite additives					

N in 3.4 mol% NaCl-33.7 mol% KCl-52.9 mol% ZnCl₂ eutectics at 250 °C and 500 °C. Hastelloys C-276 and C-22 exhibited relatively good corrosion resistance up to 500 °C, while Hastelloy N showed a higher corrosion rate (> 150 µm/yr) at 500 °C.

In general, the chemical reactivity of liquid inorganic materials increases with temperature. Corrosion problems cause poor storage performance and this section discusses these issues in achieving safe and long-term thermal storage at medium and high temperatures.

Application Generally, salt hydrates can be applied to the thermal management of building windows and walls. Weinlaeder et al. [106] investigated a sunshade system used in the office room involving several vertical slats, which were filled with commercial hydrated salt as PCMs with a melting temperature 26–30 °C. Compared with ordinary rooms with traditional shutters, using this thermal management system in an office room can reduce the indoor air temperature by up to 2 °C. Gracia et al. [107] investigated the thermophysical performance of an opaque ventilation surface, which were filled with macro-encapsulated hydrated salts with a melting temperature

of 21.5 °C. They reported that the ventilated façade supplied 2.49 MJ net energy per day.

Due to high heat storage density per unit volume, high thermal conductivity, and good stability, metals and alloys could be applied to direct steam generation applications. Blanco-Rodríguez [108] validated the feasibility of eutectic metal alloys as thermal energy storage materials by a systematic set of experiments and CFD simulations. In order to reduce heat loss, a rock wool blanket with a thickness of 0.12 m was employed to cover the entire unit. Seven multi-point sheaths (labelled 1 to 7) were placed in the axial direction to record the temperature distribution in the thermal energy storage unit, while 6 K-type thermocouples (labelled A to F) were equidistant in the radial direction.

As mentioned above, other inorganic materials have good features such as high phase change enthalpy, low vapor pressure, and relatively low prices, which make them attractive PCM options. In general, these inorganic materials and their eutectics are widely used in 200–600 °C scenarios, especially for CSP with LHS systems. The thermal energy storage system is the key to improving the efficiency, energy storage density, dispatchability, and economic sustainability of CSP plants.

The LHS heat exchanger usually consists of vertical parallel tubes with HTF inside and static PCM volumes outside. As shown in Fig. 4a, a typical heat storage configuration was designed to simulate the heat transfer of the LHS. This model can reach a maximum storage

temperature of 280 °C [109]. Figure 4b shows the LHS module with similar operating conditions (145 bar, 350 °C) to commercial CSP plants. It includes a high-pressure steam closed-loop, electric heaters, a condenser, air cooler, and pressurizer to simulate the real CSP. The PCM storage part of this pilot-scale contains a heat exchanger module equipped with steam high-pressure vertical fin tubes bundle, storage unit (low-pressure envelope of 6350 kg pure sodium nitrate), and aluminum inserts with a volume ratio of less than 19% for heat transfer enhancement [110]. This test module stores steam condensation heat during the charging period and recovers the latent heat stored in NaNO_3 to evaporate water in peak demand hours.

3.3 Heat transfer enhancement methods

Unlike metals and alloys, inorganic salts usually suffer from low thermal conductivity, so heat transfer enhancement techniques and measures need to be highlighted in order to improve thermal energy storage performance. Reinforcement technologies were studied from the aspects of materials, components and systems.

3.3.1 Material level

Common high thermal conductivity particles mainly include metallic particles and their oxides as well as carbon materials. Considering the density difference between metallic particles and inorganic salt PCMs, Tian et al. [111] mixed Li_2CO_3 - Na_2CO_3 - K_2CO_3 eutectic salt with Mg particles. It offered a thermal conductivity of 1.93 W/mK, which is 19.6–45.1% higher than that of pure

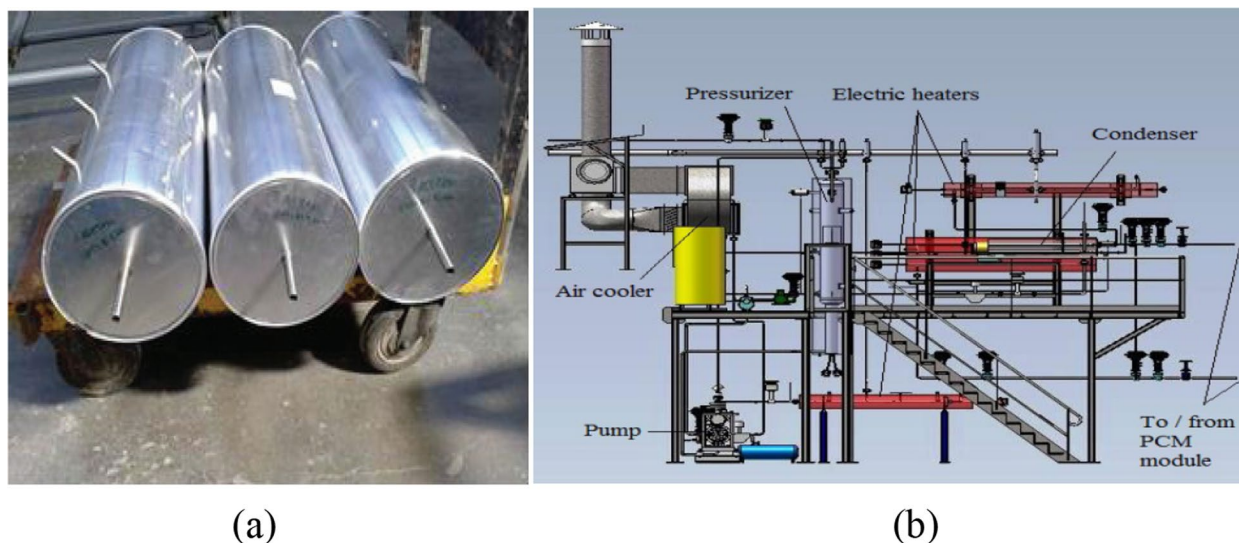


Fig. 4 Representative configurations and application of LHS based on inorganic salt PCMs: **a** “shell and tube” configuration for CSP plants and **b** the thermal energy storage module for CSP plants [109, 110]

ternary carbonate salts. Myers et al. [112] embedded CuO nanoparticles in nitrate salts and their eutectics. With the addition of 2% volume ratio of CuO, the thermal conductivity of KNO_3 and eutectics obviously improved at all the test temperatures, but the thermal conductivity of NaNO_3 only increased under 150 °C.

Carbon materials are used to improve the thermal conductivity of a PCM due to their high thermal conductivity, high corrosion resistance, and relatively low density. Pincemin et al. [113] investigated the effect of dispersing different amounts of graphite flake and expanded graphite powder in Solar Salt on the thermal conductivity of the composites. They showed that when 10wt% graphite was incorporated, a larger particle size could lead to a low thermal conductivity, while for 20 wt% graphite, the larger particle size facilitated the creation of a conductive network. Tao et al. [114] studied the effects of embedding single-walled carbon nanotubes (SWCNTs), MWCNT, carbon 60, and graphene in the Li_2CO_3 - K_2CO_3 eutectic on its thermal conductivity. Except for carbon 60, the others enhanced the thermal conductivity of the mixtures to varying degrees, and the effect of adding SWCNT was the best. Table 7 summarizes the thermal conductivity of PCMs embedded in highly thermal conductive particles.

Embedding porous media with high thermal conductivity and large specific surface areas are another effective way to enhance the heat transfer of PCMs. Jiang et al. [122] developed a novel form-stable nitrate/calcium silicate PCM by cold compression and sintering. Its thermal conductivity is about 1.177 W/(mK). Huang et al. [123] designed a LiNO_3 -KCl/EG composite PCM with 10–30 wt% EG, and found that thermal conductivity of the composite increased with the density when the EG mass fraction was 20%. Table 8 summarizes the thermal conductivity of some composite PCMs embedded in porous media.

Open-cell foam materials can solve the two problems: the thermal conductivity anisotropy caused by the discontinuous layered structure of EG, and phase separation after several cycles caused by the density difference of the composite PCMs. Zhao et al. [130] numerically investigated the undercooling effect of metal foams during solid–liquid phase change by the phase field method. Zhao and Wu [131] experimentally studied the effect of EG/metal foams on the melting of NaNO_3 . Both EG and metal foams can significantly enhance heat transfer of NaNO_3 , and metal foams showed a better overall performance compared to EG.

3.3.2 Component level

Typically, heat transfer is limited not only to the material level (Section 3.3.1), but also to the component

level [132]. Using fins or enhanced heat transfer tubes is a potential solution. Common fins include longitudinal fins, annular fins, and pin fins. Steinmann and Tamme [133] formed a sandwich structure by combining foils with PCMs. Tao and He [134] designed locally reinforced finned tubes to improve the uniformity of the melting process. Although local fins can improve the uniformity of the LHS performance, large fins will destroy the uniformity again, so it is necessary to appropriately select the fin size. Sciacovelli et al. [135] optimized Y-shaped fins through the combination of CFD modeling and response surface methods with the system efficiency increased by 24% as shown in Fig. 5a. It was found a shorter operating time requires a larger angle Y-shaped fin. Conversely, a smaller angle is required.

However, the optimization method described above is limited to low dimensions. Topology optimization helps to perform highly discrete optimization, which has been applied to heat transfer problem field. Pizzolato et al. [137] performed a topology optimization on heat transfer fins with natural convection. The optimized design for melting was fundamentally different from fins neglecting fluid flows. Zhao et al. [136] discussed the effect of topology optimization on the fin distribution in a heat storage unit by experimentally comparing the optimized fin with the regular one, as shown in Fig. 5b.

Adding heat pipes can also effectively improve the heat transfer of PCMs at medium- or high-temperatures. Shabgard et al. [138] added heat pipes to a LHS application, and demonstrated that the heat pipe enhanced the thermal performance. However, due to the volume expansion of PCMs, the shell may bear a certain pressure. There is a risk of cracking. Steinmann and Tamme [133] chose cylindrical capsules to wrap eutectic $\text{NaNO}_3/\text{KNO}_3$. They found that the initial gas volume determined the pressure inside the capsules. When the initial gas volume was less than 20%, the pressure significantly increased after melting.

3.3.3 System level

Heat transfer enhancement at the system level is mainly concentrated in cascaded latent heat storage (CLHS), which is an efficient and economical method to store thermal energy arranging multilayer phase change materials according to melting temperatures [20]. The idea of multiple PCMs was proposed, and then it was studied by numerical simulation [139] or experimental studies [140]. It showed that compared with a single PCM system, multiple PCMs system offer a higher heat transfer rate. Xu and Zhao [141] conducted an optimization for a CLHS system and obtained the analytical solutions of the optimum temperature for any stage. Most of the

Table 7 Thermal conductivity of some composite PCMs embedded in highly thermal conductive particles

PCMs Material	<i>k</i> W/mK	Additives Material	Size	Amount	Thermal conductivity W/mK	Refs
32.2% Li ₂ CO ₃ - 33.2% Na ₂ CO ₃ - 34.5% K ₂ CO ₃ , wt%	1.33	Mg	100–200 meshes	0.1 wt%	1.59	[111]
Solar Salt	0.78	MgO	-	2 wt%	1.93	[115]
				0.125 wt%	0.85	
				0.25 wt%	0.88	
				0.5 wt%	0.85	
				1.0 wt%	0.86	
32.1% Li ₂ CO ₃ - 33.4% Na ₂ CO ₃ - 34.5% K ₂ CO ₃ , wt%	1.67	Al ₂ O ₃	20 nm	2 wt%	0.87	[116]
				1.0 wt%	2.06	
				1.0 wt%	2.15	
32.1% Li ₂ CO ₃ - 33.4% Na ₂ CO ₃ -34.5% K ₂ CO ₃ , wt%	1.67	T-ZnOw	-	80 nm	2.19	[117]
				0.2 wt%	2.06	
				0.4 wt%	2.64	
				0.8 wt%	3.36	
				1.0 wt%	3.89	
LiNO ₃	0.51	Graphite	85 μm	1.4 wt%	4.48	[118]
				2.0 wt%	4.36	
				5 wt%	0.87	
				10 wt%	1.19	
				15 wt%	1.52	
Na ₂ CO ₃ /MgO	0.75 @40 °C 0.83 @75 °C 0.88 @120 °C	MWCNT	<i>l</i> = 10 μm <i>d</i> = 11 nm	20 wt%	1.86	[119]
				0.1 wt%	0.90 @40 °C	
					1.03 @75 °C	
					1.14 @120 °C	
				0.3 wt%	1.05 @40 °C	
					1.20 @75 °C	
					1.65 @120 °C	
				0.5 wt%	1.13 @40 °C	
49% Na ₂ CO ₃ - 51% Li ₂ CO ₃ , wt%	0.69	MgO MgO+graphene Cu+MgO	-	-	1.30 @75 °C	[120]
					1.49 @120 °C	
					0.87	
Solar Salt	0.41 @300 °C	MWCNT	<i>l</i> = 3–15 μm <i>d</i> = 12–15 nm	0.3 wt%	1.24	[121]
				0.4 wt%	1.34	
					1.62	
					0.78	

studies on medium-temperature CLHS were theoretical optimization and numerical simulation, while there are few experimental studies. Mawire et al. [142] studied the charging process of the CLHS system (as shown in Fig. 6a) which used eutectic solder (Sn63/Pb37), erythritol and adipic acid as multiple PCMs. Zhao et al. [143] established a CLHS system to study the comprehensive thermal performance. They chose three nitrate composites with melting temperatures of 125.58 °C, 224.42 °C, and 306.36 °C as multiple PCMs. They found that higher HTF inner temperatures and more stages could enhance

the system in entransy storage efficiency, exergy storage efficiency, and energy storage efficiency. Yuan et al. [144] created a three-stage CLHS system (as shown in Fig. 6b) and used carbonate composites as PCMs, whose melting temperatures were 422 °C, 484 °C and 499.9 °C, respectively and solved the problem of incomplete melting.

3.4 Summary

LHS based on PCMs with high energy density is an attractive energy storage option. There are many types of PCMs. They can be divided into two categories: organic

Table 8 Thermal conductivity of some composite PCMs embedded in porous media

PCMs Material	<i>k</i> W/mK	Porous media Material	Amount	Thermal conductivity of form- stable composites W/mK		Refs
NaNO ₃	0.50	Expanded Perlite	10 wt%	0.84/1.14	@25/300 °C	[124]
			20 wt%	0.67/0.81	@25/300 °C	
			40 wt%	0.54/0.62	@25/300 °C	
			60 wt%	0.42/0.57	@25/300 °C	
58.1% LiNO ₃ - 41.9% KCl, wt%	0.93	Expanded graphite	18.3 wt%	5.59		[125]
49% LiNO ₃ - 51% NaNO ₃ , wt%	0.84		18.2 wt%	6.61		
87% LiNO ₃ - 13% NaCl, wt%	0.80		24.2 wt%	4.71		
Solar Salt	2.27	Expanded graphite	0.5 wt%	2.88		[126]
			1 wt%	3.34		
			1.5 wt%	4.14		
			2 wt%	4.88		
40% LiNO ₃ - 60% KNO ₃ , wt%	1.03 @5 MPa	Expanded graphite	5 wt%	4.19	@5 MPa	[127]
	1.10 @15 MPa			5.34	@15 MPa	
	1.30 @25 MPa			5.53	@25 MPa	
				1.36	@5 MPa	
			30 wt%	12.41	@15 MPa	
				13.04	@25 MPa	
				15.42	@5 MPa	
				16.29	@15 MPa	
53.4% NaCl - 15.0% CaCl ₂ -31.6% MgCl ₂ , mol%	1.17	Expanded graphite	0.5 wt%	1.58		[128]
			0.7 wt%	1.64		
			1 wt%	1.64		
			3 wt%	1.68		
			5 wt%	2.08		
31.5% MgCl ₂ - 68.5% KCl, mol%	0.41	Expanded graphite	15 wt%	4.92		[129]

compounds and inorganic materials. This section summarizes the thermophysical properties, cycle stability, thermal stability, corrosion, and application of various PCMs according to the second classification method.

Organic compounds and hydrated salts are widely used in low- and medium-temperature conditions because of their good corrosion resistance and favorable characteristics of congruent melting and self-nucleation, but low melting temperatures and latent heat limit their development as high-temperature PCMs. In contrast, inorganic salts, and metals, have relatively higher melting temperatures and phase change enthalpies, which make them attract widespread attention. However, as the operating temperature increases, the chemical reactivity of liquid metals and inorganic salts becomes active, which may lead to changes in composition or structure, or even decomposition, especially for inorganic salts and their eutectics. It is necessary to study the cycling and thermal stability. In addition, inorganic salts and metals also suffer from corrosion problems, but this is not a problem for

organic materials. The study of corrosion can also effectively improve LHS performance.

Generally, PCMs have low thermal conductivity, which is a great challenge for the application of LHS. This section summarizes the main methods and techniques to enhance heat transfer at the material, component, and system levels respectively.

The research on new PCMs and innovative technologies is still developing. For example, the research on the technology of photosensitive materials to control the occurrence of phase change. The development of low-cost, excellent thermodynamic properties, stable, and safe PCMs are the needs of the industrial society and the key to large-scale application of LHS technologies. Large-scale LHS systems with mature technologies and low costs will promote the realization of carbon peak and carbon neutrality.

4 Thermochemical energy storage

4.1 Classifications by temperature range

Temperature range is an important parameter for TCES, which determines its application field. Different

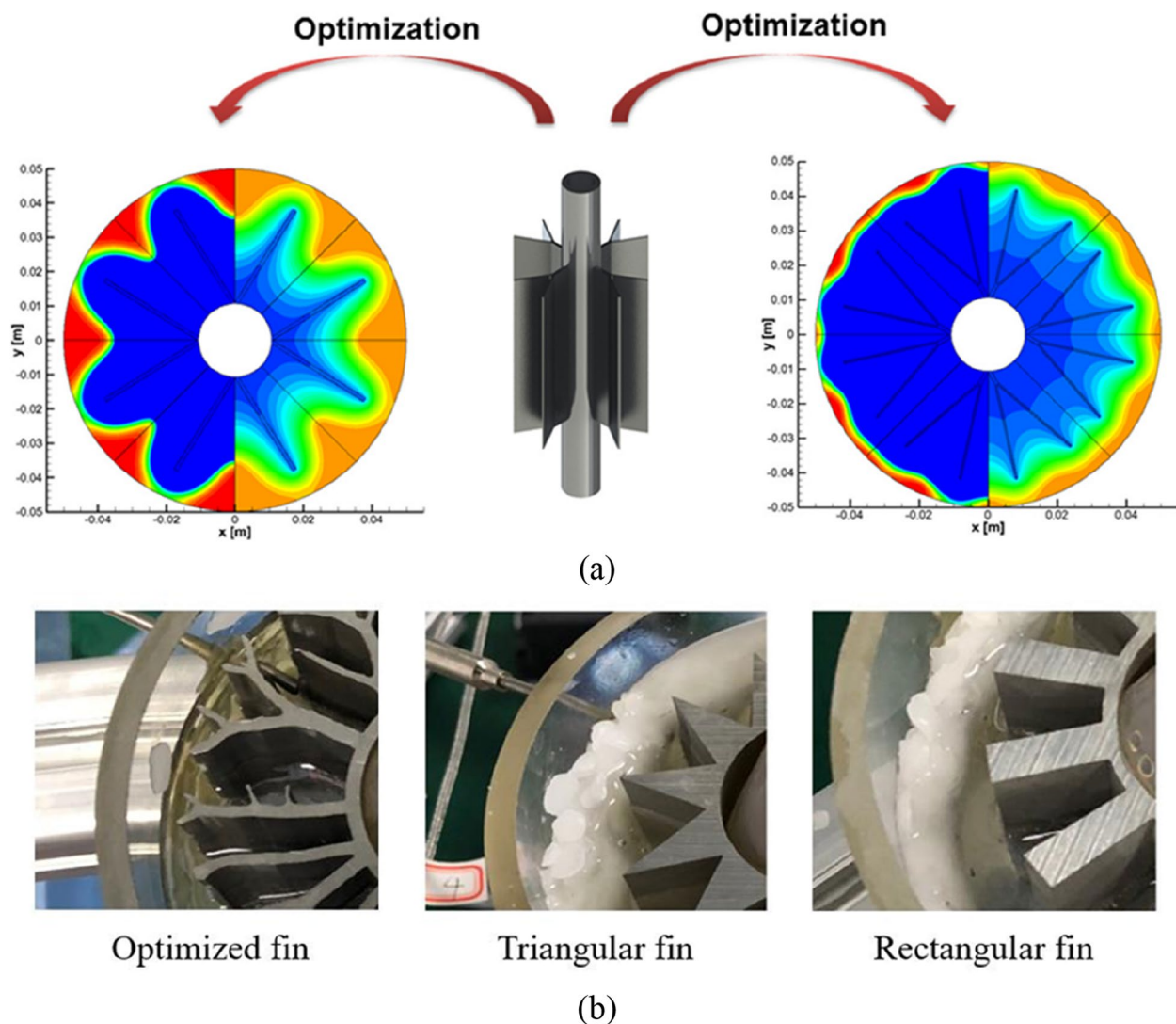


Fig. 5 Optimal fin design, and corresponding temperature and liquid fraction distributions: **a** response surface method and **c** experimental validation of topology optimization [135, 136]

reversible reactions provide various targeted working temperature ranges at solar inputs and heat outputs. The temperature zone for TCES is mainly divided into the low-temperature zone ($< 150\text{ }^{\circ}\text{C}$), medium-temperature zone ($150\text{--}350\text{ }^{\circ}\text{C}$), and high-temperature zone ($> 350\text{ }^{\circ}\text{C}$). The wide temperature ranges can meet different demands of industry and life for heat sources.

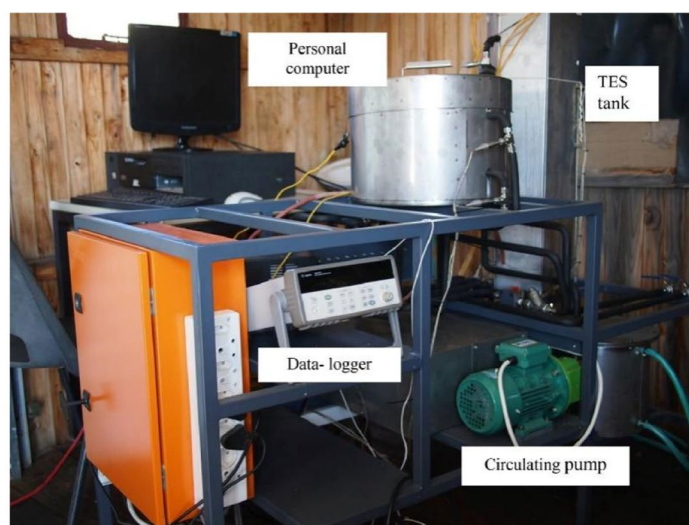
TCES materials in the low-temperature zone are mainly sorption materials, including Silica gels, Zeolites, Zeotypes (AlPOs and SAPOs) and Metalorganic Frameworks (MOFs) [145], which are not emphatically introduced in this paper. TCES materials in the medium-temperature zone include some metal hydrides (such as Mg_2NiH_4 [146] and MgH_2 [147]) and metal hydroxides (such as $\text{Mg}(\text{OH})_2$ [148]). In the medium- and high-temperature zone, metal hydroxides (such as $\text{Ca}(\text{OH})_2$ [149]) and

metallic carbonates (such as MgCO_3 [150]) are mainly included. More TCES materials are used in the high-temperature zone, including some metal hydrides (such as NaMgH_3 [151], TiH_2 [152], CaH_2 [153] and NaH [154]), metallic carbonates (such as CaCO_3 [155], SrCO_3 [156], BaCO_3 [157] and PbCO_3 [158]) and metal oxides (such as Co_3O_4 [159], BaO_2 [160], Mn_2O_3 [161] and CuO [162]). The present review is mainly focused on solid–gas TCES reactions mentioned above.

4.2 Various thermochemical heat storage systems

4.2.1 Metal hydrides

Metal hydrides are promising materials for TCES applications. During the energy storage process, metal hydrides (MH) decompose and absorb heat, resulting in the formation of gaseous H_2 and the solid metals.



(a)



(b)

Fig. 6 Representative lab-scale experimental CLHS systems: **a** Mawire et al.'s CLHS system, and **b** Yuan et al.'s CLHS system [142, 144]

During the heat release process, the backward reaction is conducted and release heat that is stored in the chemical bonds of the reaction products [163]. The reversible reactions of metal hydrides have been studied for hydrogen storage applications. The main advantage of metal hydride systems is the high energy storage density, and the main prominent metal hydrides for TCES implemented in CSP plants are shown in Table 9 [164].

Although there are many TCES systems mentioned above, many of them have low technology development. Mg-based hydrides are the most studied for this application, but it is still in the initial studies, such as the kinetic and thermal conductivity problems. Meanwhile, the high operating temperature and pressure also limits its large-scale application. Paskevicius et al. [165] used

supercritical water as the heat transfer fluid to build a prototype. The hydrogen uptake process was slower than H_2 release, which resulted in slower absorption kinetics. Shen et al. [166] established a two-dimensional mathematical model for the heat releasing process of the Mg/MgH₂ system. They found that the heat removal ability was obviously enhanced and the bed temperature had an obvious drop which led to a faster reaction rate. Adding metal foams can achieve a 40% reduction of the reaction time and 60% promotion of the exothermic power. Chaise et al. [167] improved the thermal conductivity of the Mg/MgH₂ system up to $8 \text{ W m}^{-1} \text{ K}^{-1}$ by doping 10% of expanded natural graphite. In addition, Rönnebro et al. [168] also increased thermal conductivity of titanium hydride from 10 to $20 \text{ W m}^{-1} \text{ K}^{-1}$ at 500°C . The

Table 9 Metal hydrides candidates for TCES [164]

TCES systems	Temperature (°C)	Energy storage density (kJ/kg)	Energy storage density (MJ/m ³)
NaH/NaMgH ₃	430–585	1463	2134.8
Mg/MgH ₂	300–400	2389	3463.2
Ti/TiH _{1.7}	700–1000	851	3243.6
Ca/CaH ₂	> 1000	4198	7016.4
Mg-Fe/Mg ₂ FeH ₆	350–550	1776.5	4867.2

researchers have done some efforts to settle the kinetic and thermal conductivity problems, but it is still in the initial studies. In addition, Ca/CaH₂ system has drawn much attention because of the advantage of low material cost. In fact, the Australian company EMC has developed a TCS reactor with 50 kg of CaH₂ coupled to a Stirling engine [163].

Above all, metal hydrides have advantages of high energy density and wide temperature ranges, which are hoped to be applied in concentrator configurations in the market. In addition, several energy storage units have been investigated. However, this technology requires high temperature and pressure for operation. Furthermore, the separation and storage of H₂ is also an important problem, which poses limitations on the development of this technology. The reaction kinetics, cyclic stability, and heat and mass transfer in the reactor still needs to be further studied.

4.2.2 Metal hydroxides

Material selection The hydroxide system is promising due to the large energy density and safe reactive gas (i.e. vapor). The study of alkaline earth metal oxides for thermal energy storage started in the 1970s. Wentworth [169] proposed that alkaline earth metal hydroxides could be used for thermal storage, such as Ca(OH)₂, Mg(OH)₂, Sr(OH)₂ and Ba(OH)₂, etc. The decomposition temperatures of these materials are concentrated between 300 and 700 °C. Among them, Ca(OH)₂ and Mg(OH)₂ have drawn the most attention due to their low cost and non-toxicity. Their heat storage densities are 104.4 kJ/mol and 81 kJ/mol, respectively, and their specific heat capacity (*c_p*) at 300 °C is 1.3 kJ/(kg·K) [170] and 1.7 kJ/(kg·K), respectively. Their low cost and large heat storage density make them promising for large-scale industrial applications. Sr(OH)₂ and Ba(OH)₂ will partially evaporate during the dehydration process due to their low melting points. Doping MgO can improve this phenomenon [171] but it will greatly reduce the heat storage density. Therefore, they are not convenient for heat storage applications.

Since Ca(OH)₂ and Mg(OH)₂ usually exist in the forms of power in the energy storage process, the low thermal conductivity of these materials limited their applications [172]. For example, the thermal conductivity of pure calcium hydroxide particles in the reaction bed was measured by experiments to be 0.12 W/(m·K) [173]. Some studies have been conducted to improve the heat transfer properties. Wang et al. [52] simulated a porous channel in the reactor, and simulated results showed that the thermal conductivity was significantly enhanced. Kariya et al. [174] used a silicon carbide—diesel particle filter (SiC-DPF) to support Ca(OH)₂ for application. The merits of SiC-DPF include chemical stability, low cost, large pore volume and high thermal conductivity of 90 W/(m·K), which make Ca(OH)₂ get over the low thermal conductivity and vapor diffusivity through TCES materials at larger scale. Shkatulov et al. used KNO₃ as the dopant to make the dehydration temperature decrease by 35 °C. In addition, fast rehydration of the doped CaO is observed at the temperature range of 290–360 °C. Huang et al. prepared the hexagonal boron nitride (HBN)-doped Ca(OH)₂ by ultrasonic and mechanical agitation. The thermal conductivity of Ca(OH)₂ was improved by 22.9% at 300 °C. Since the similarity of properties between Ca(OH)₂ and Mg(OH)₂, these methods can also be used in the thermochemical energy storage of Mg(OH)₂. These efforts promote the application of hydroxide-based thermochemical energy storage.

Cycling stability The cycling stability of Ca(OH)₂/CaO and Mg(OH)₂/MgO was investigated by Ervin [175] in 1977. Ca(OH)₂/CaO was subjected to 211 cycling experiments, and the results showed that the hydration ratio of CaO was maintained at about 95%. It indicated that Ca(OH)₂/CaO has good cycling stability. However, the reaction rate gradually decreased and remained stable from the 190th cycle. Rosemary et al. [176] also tested the dehydration/hydration cycles up to 1171 times. The agglomeration of Ca(OH)₂ particles occurs especially during the cycling of CaO/Ca(OH)₂ is the worsening of the reaction material [177], since the fast discharging demands a high HTF velocity leads to a high pressure drop in the reaction bed due to the low permeability of the power bed. A homogeneous flow of the reaction and heat transfer fluid gases through the fixed reaction bed is not possible when severe agglomeration occurs. Roßkopf et al. [178] inhibited the agglomeration of thermal storage systems by adding hydrophilic SiO₂ nanoparticles to Ca(OH)₂. Afflerbach [179] obtained mechanically strong particles by covering the shell with porous ceramics outside of the thermal storage material. Xia [180] prepared stable granular thermal storage materials by doping with

sodium carboxymethyl cellulose. The structural integrity was maintained after several cycles. In addition to material agglomeration, CO₂ in the air is another important influence on the cycling stability of Ca(OH)₂. Anthony [181] pointed out that CaO with a larger surface area is more likely to capture CO₂. Yan [171] compared the effect of CO₂ in the air on the heat storage and exothermic process of Ca(OH)₂. It was found that the heat storage material is more susceptible to failure by CO₂ due to the presence of water vapor in the hydration process.

For Mg(OH)₂/MgO, Ervin [175] conducted 500 cycles and the hydration ratio of MgO decreased from 95 to 60%-70% in the first 40 cycles. However, the hydration ratio of MgO tended to be stable in the next 460 cycles. Kato found that the cycling stability of the MgO hydration process is closely related to the particle size and purity of the raw material [182]. When the particle size is too large, the emerging product layer hinders the gas transport and thereby decreases the effective reaction rate, which makes it difficult for the reaction to continue. Also, the degradation of the surface activity of the porous structure inside MgO also leads to a decrease in hydration conversion. In addition, similar to the property of Ca(OH)₂, Mg(OH)₂ can also be affected by CO₂ and lose dehydration/hydration activities. However, the effect of CO₂ on the Mg(OH)₂/MgO system is smaller than that of Ca(OH)₂ [171]. In a word, the material agglomeration is an important influence on the cycling stability for Ca(OH)₂/CaO and Mg(OH)₂/MgO system.

Reaction kinetics The interaction effects between the reaction kinetics and heat transfer are complicated for a TCES process. Therefore, it is important to obtain the clear reaction kinetic parameters. Matsuda [183] studied the decomposition kinetics of Ca(OH)₂ from the 1980s and obtained the governing equations. The common format which can be used to describe reaction kinetics of both Ca(OH)₂ and Mg(OH)₂ are shown as follows [184–186]:

$$\frac{dX}{dt} = k(T) \cdot h(p, p_{eq}, T, T_{eq}) \cdot f(X) \quad (2)$$

Particularly, $k(T)$ is a function accounting for the temperature, which is typically described by the Arrhenius equation [149]:

$$k(T) = A \cdot \exp\left(-\frac{E}{RT}\right) \quad (3)$$

where A refers to the pre-exponential factor, E refers to the activation energy, and R refers to the universal gas constant.

In Eq. (2), the function $h(p, p_{eq}, T, T_{eq})$ depends on the steam pressure p , the equilibrium pressure p_{eq} , the temperature T , and the equilibrium temperature T_{eq} . They describe the influence of distance to the thermodynamic equilibrium. $f(X)$ is a function of the conversion describing the growth mechanism during the reaction. The kinetic models of $f(X)$ are summarized in Table 10 [187]. The whole kinetic models of Ca(OH)₂ dehydration and CaO hydration are shown in Table 11. It exhibits that the kinetics parameters are also affected by other conditions, such as material properties, reaction types and reaction processes.

The equilibrium temperature and partial pressure of water vapor of the reaction satisfy [149]:

$$\ln\left(\frac{p_{eq}}{10^5}\right) = -\frac{12845}{T_{eq}} + 16.508 \quad (4)$$

where p_{eq} and T_{eq} refer to the equilibrium pressure and temperature, respectively. Alternatively, Criado et al. [188] compared different equilibrium equations and found their equilibrium temperatures were very close to a predicted model, which is shown as follows:

$$P_{eq} = 2.30 \times 10^8 \exp\left(-\frac{11607}{T_{eq}}\right) \quad (5)$$

Many researchers have studied the effect of doping modified materials on the reaction kinetics of Ca(OH)₂. Murthy [191] found the addition of Ni(OH)₂, Zn(OH)₂, and Al(OH)₃ cause non-uniform nucleation. On the other hand, these doped materials play a catalytic-like role and reduce the reaction activation energy. Li et al. [192]

Table 10 The kinetic models of $f(X)$ [187]

Reaction model	Code	$f(X)$
Power law	P4	$4X^{\frac{3}{4}}$
Power law	P3	$3X^{\frac{2}{3}}$
Power law	P2	$2X^{\frac{1}{2}}$
Power law	P2/3	$\frac{2}{3}X^{-\frac{1}{3}}$
One-dimensional diffusion	D1	$\frac{1}{2}X^{-1}$
Mampel (first order)	F1	$1 - X$
Avrami–Erofeev	A4	$4(1 - X)[- \ln(1 - X)]^{\frac{3}{4}}$
Avrami–Erofeev	A3	$3(1 - X)[- \ln(1 - X)]^{\frac{2}{3}}$
Avrami–Erofeev	A2	$2(1 - X)[- \ln(1 - X)]^{\frac{1}{2}}$
Three-dimensional diffusion	D3	$\frac{3}{2}(1 - X)^{\frac{2}{3}}[1 - \ln(1 - X)^{\frac{1}{3}}]^{-1}$
Contracting sphere	R3	$3(1 - X)^{\frac{2}{3}}$
Contracting cylinder	R2	$2(1 - X)^{\frac{1}{2}}$
Two-dimensional diffusion	D2	$[- \ln(1 - X)]^{-1}$

Table 11 The whole kinetic models of Ca(OH)_2 dehydration and CaO hydration

$A(\text{s}^{-1})$	$E(\text{kJ/mol})$	$h(p, p_{\text{eq}}, T, T_{\text{eq}})$	$f(X)$	Reaction stage	Ref
15	0	$\frac{T_{\text{eq}} - T}{T_{\text{eq}}}$	$1 - X$	CaO hydration	[184]
$\frac{3.5 \times 10^{-4}}{d_p [\mu\text{m}]}$	-59.4	$\frac{p_{\text{H}_2\text{O}} - p_{\text{eq}}}{p_{\text{total}}}$	$3(1 - X)^{\frac{2}{3}}$	CaO hydration	[188]
390.827	87.46	$[\max(\frac{1}{p_{\text{H}_2\text{O}}/p_{\text{eq}}} - 1)]^{3.43}$	$1 - X$	CaO hydration	[189, 190]
13,945	89,486	$(\frac{p_{\text{H}_2\text{O}}}{p_{\text{eq}}} - 1)^{0.83}$	$3(1 - X)[- \ln(1 - X)]^{\frac{2}{3}}$	CaO hydration	[149]
$1.0004 \cdot 10^{-34}$	-443.427	$(\frac{p}{10^5 p_a})^6$	$1 - X$	CaO hydration	[149]
449,974	91.282	$[1 - \min(\frac{1}{p_{\text{H}_2\text{O}}/p_{\text{eq}}})]^{3.47}$	X	Ca(OH)_2 dehydration	[189]
$1.9425 \cdot 10^{12}$	187.88	$(1 - \frac{p}{p_{\text{eq}}})^3$	$1 - X$	Ca(OH)_2 dehydration ($X < 0.2$)	[149]
$8.9588 \cdot 10^9$	162.62	$(1 - \frac{p}{p_{\text{eq}}})^3$	$2(1 - X)^{\frac{1}{2}}$	Ca(OH)_2 dehydration ($X > 0.2$)	[149]

lowered the dehydration temperature of Ca(OH)_2 by doping with nitrate $\text{ZrO}(\text{NO}_3)_2$, which reduced the activation energy of the dehydration reaction from 176 kJ/mol to 130 kJ/mol.

For the reaction kinetics of $\text{MgO}/\text{Mg(OH)}_2$, The effect of water vapor pressure and temperature on the rate of the hydration reaction was quantitatively studied by Kato [193]. The kinetic process is divided into three stages. The first stage is the structural water stage, where water vapor pressure and temperature have almost no effect. The second stage is the adsorbed water stage. This stage is mainly influenced by the reaction temperature and has little relationship with water vapor pressure. The third stage is the chemical reaction stage, which is influenced by the combined effect of water vapor pressure and temperature. Pan [194] investigated the relationship between the dehydration temperature of MgO and the hydration conversion rate. When the vapor pressure is constant, increasing the hydration temperature can improve the efficiency of the heat storage system. Compared with the previous formula Eq. (2), the reaction kinetic of Mg(OH)_2 is shown as follows [195]:

$$\frac{dX}{dt} = 1.51 \times 10^8 \exp\left(-\frac{1.33422 \times 10^5}{RT}\right) \cdot (1 - X)^{0.67} \quad (6)$$

A is 1.51×10^8 , and $f(X)$ is $(1 - X)^{0.67}$.

Systematic studies were carried out to reduce the decomposition temperature of hydroxide and increase the decomposition rate of hydroxide. The decomposition reaction of pure Mg(OH)_2 can only occur above 277 °C [194]. In order to enable the application of Mg(OH)_2 to cover the 200–300 °C waste heat recovery system, Ryn et al. [196] first proposed adding LiCl , NaCl , and CaCl_2 to Mg(OH)_2 to reduce the decomposition temperature

of Mg(OH)_2 . The lowest decomposition temperature of the composite was 233 °C after adding 6.8 wt% LiCl and 267 °C after adding 6.4 wt% NaCl . The addition of salts reduces the reaction temperature of Mg(OH)_2 mainly because the additives reduce the activation energy of Mg(OH)_2 decomposition. In addition, the strong water adsorption capacity of salts also plays a role. Alexandr et al. [197] found that doping expanded vermiculite and Mg(OH)_2 could reduce the decomposition temperature by about 50 °C, but the reason for lowering Mg(OH)_2 decomposition temperature is not clear.

In general, there many studies on the influence of gas pressure and temperature on the kinetics, but the reasons of the low kinetics situations are not clear. The method of doping materials can improve the reaction kinetics properties. However, study on the mechanism of doping is insufficient.

Reactor applications Many current research on reactors for TCES systems is focused on fixed bed reactors and mobile bed reactors. Schmidt [198] conducted an experimental study of a $\text{CaO}/\text{Ca(OH)}_2$ fixed bed with approximately 20 kg of Ca(OH)_2 , as shown in Fig. 7. It was found that the input and output power of the fixed bed is limited by the heat transfer fluid and the heat transfer of the bed. Roßkopf et al. [178] improved the fluidity of Ca(OH)_2 and reduced the agglomeration of powder by doping nano SiO_2 in Ca(OH)_2 . Using a similar method, Schmidt et al. [199] established a 10 kW/100kWh mobile bed reactor. These mobile bed reactor systems are usually able to achieve continuous and stable heat exchange. Yan et al. [200] experimentally showed that the higher the vacuum level of the system, the faster the progress of the exothermic reaction. The presence of air in the reactor prevents the outflow of water vapor during the heat



Fig. 7 Reactor with 20 kg of Ca(OH)_2 [198]

storage process and prevents water vapor from entering the bed for the reaction during the exothermic process. In addition to the partial strengthening of materials, optimization of the reactor structure can also improve the heat storage and exothermic performance of the reactor. Chen et al. [201] improved the heat transfer of a fixed bed reactor by means of topology optimization shown in Fig. 8.

In addition, some investigations also focus on fluidized bed reactors. Rougé et al. [202] conducted CaO/

Ca(OH)_2 TCES by using a fluidized bed reactor. The experimental results during dynamic and steady state periods were fitted to a KL reactor bubbling bed model, using kinetic parameters from thermogravimetric studies and a single crossflow factor. Pardo et al. [203] used the sample of 70 wt% Al_2O_3 /30 wt% Ca(OH)_2 in a fluidized bed reactor to perform Ca(OH)_2 dehydration and CaO hydration reactions. The mean energy density obtained is 60 kWh/m^3 , which amounts to a promising energy density of 156 kWh/m^3 of bulk Ca(OH)_2 . Criado et al. [204] experimentally investigated the hydration and dehydration reactors in a $5.5 \text{ kW}_{\text{th}}$ batch fluidized bed reactor and built a standard 1D bubbling reactor model. They found that the hydration reaction is mainly controlled by the slow kinetics of the CaO material tested while significant emulsion-bubble mass-transfer resistances were identified during dehydration due to the much faster dehydration kinetics. Bian et al. [205] studied the effects of critical factors such as particle size, fluidization number, steam concentration in hydration, initial temperature, cycle number and type of carrier gas on heat release performance of CaO in the fluidized bed reactor. It provides an important reference for the development of CaO/ Ca(OH)_2 system using the fluidized bed reactor. Risthaus et al. [206] developed a novel reactor concept based on a plow share mixer for fluidization of CaO/ Ca(OH)_2 system, demonstrated its feasibility and investigated its heat and mass transfer performance, as shown in Fig. 9.

4.2.3 Metallic carbonates

Metallic carbonates have broad application prospects for storing thermal energy because of their high energy storage density, low operating pressure, and non-corrosive property. In addition, most of them are operated at high temperatures, which is appropriate for high-temperature CSP and industrial applications. The endothermic

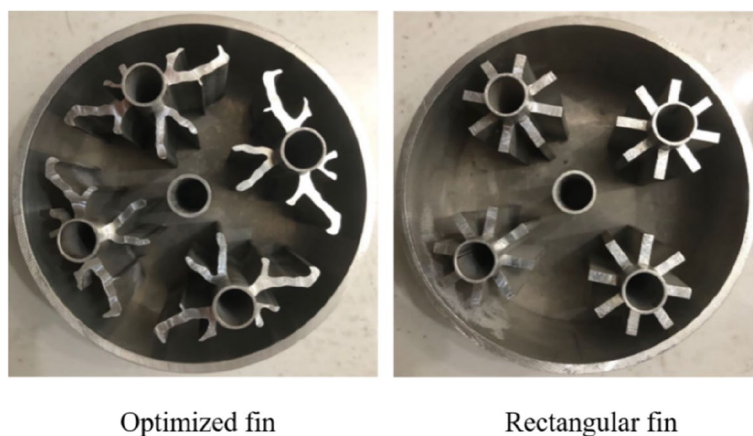


Fig. 8 Fins before and after topological optimization [201]

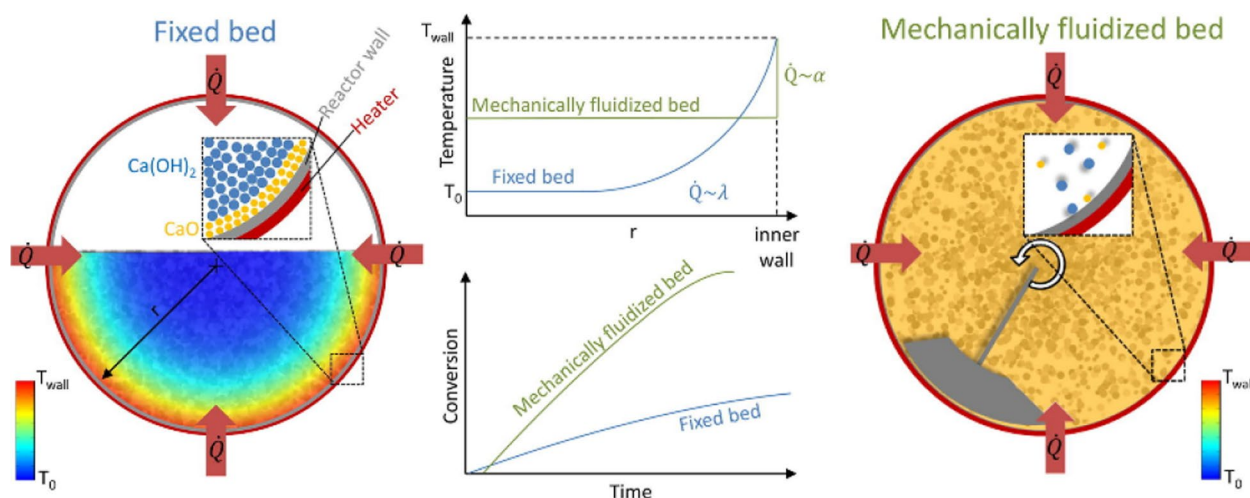


Fig. 9 The novel reactor concept based on a plow share mixer for fluidization of CaO/Ca(OH)₂ system [206]

calcination stage and exothermic carbonation stage are used to store and release the thermal energy. At the calcination stage, metallic carbonates decompose and absorb heat at relatively high temperatures. At the carbonation stage, metallic oxides react with CO₂, and heat is released at relatively low temperatures. Both reaction products are stored until the thermal energy is required.

Material research The equilibrium temperature of MgO/MgCO₃ TCES system under the ambient pressure is 398 °C, which increase with rising pressure [207]. Its reaction temperature range is from 320 °C to 431 °C at atmospheric pressure. The specific heat capacity of MgCO₃ is about 0.84 kJ/(kg·K).

The reversible carbonation/calcination of the CaO/CaCO₃ energy storage cycle stands out due to its characteristics of high energy storage density, high reaction temperature, and low cost of materials. The Calcium Looping (CaL) process was proposed to be used for TCES in the late 1970s [208]. The specific heat capacity of CaCO₃ is 0.90 kJ/(kg·K). In the decomposition stage, CaCO₃ decomposes and absorbs heat at high temperatures. In the carbonation stage, CaO reacts with CO₂ and releases heat at comparatively low temperatures (e.g., 650 °C, pure CO₂) [209]. Many studies are focused on the reaction properties of these two materials, mainly including cyclic stability and reaction kinetics.

Cycling stability The cyclic stability of CaCO₃ is the primary problem limiting its large-scale industrial applications. The unsolved significant reactivity declines of CaO during repeated cycles occurs due to the sintering of CaO [210]. Natural CaCO₃ minerals such as limestone suffer

from the decline of heat storage performance attributed to this sintering phenomenon, and the rapid formation of CaCO₃ product layer covers and blocks the pores on the surface of CaO [211, 212]. Tian et al. [213] investigated the sintering mechanism of CaO materials during TCES process as shown in Fig. 10. They found that the carbonation process was the dominant stage where sintering occurred. The decrease of diffusion activation energy and increase of the diffusion pre-exponential factor accelerated the sintering process, which should be prohibited in the future.

Moreover, some metal salts were applied to enhance the cyclic stability of the CaO/CaCO₃ heat storage system. Jimenez et al. [214] used limestone and dolomite as raw materials, acetic acid as the organic acid, and MgO as the dopant to prepare CaMg₅₀Ac composite materials by the sol–gel method. The effective conversion of modified materials after the 1st cycle ($X_{ef,1}$) was 0.78 and it only decreased 12.8% after 30 cycles. Khosa et al. [210] also mechanically mixed limestone and nano-SiO₂. The $X_{ef,1}$ was 0.34 and a 44.1% decrease of effective conversion occurred after 45 cycles. Khosa et al. [215] synthesized CaCO₃/ZnO composites by the mechanical mixing method. The $X_{ef,1}$ was 0.82 and a 50% decrease of effective conversion occurred after 50 cycles.

Al₂O₃ is reported as an effective dopant to improve the cyclic stability of calcium-based materials. Benitez-Guerrero et al. [216] mechanically mixed limestone and nano-Al₂O₃ and the $X_{ef,1}$ of the modified material was 0.71. Its effective conversion dropped to 0.6 after 20 cycles and the $X_{ef,20}$ of limestone was 0.16. Furthermore, Møller et al. [217] investigated the energy storage performance

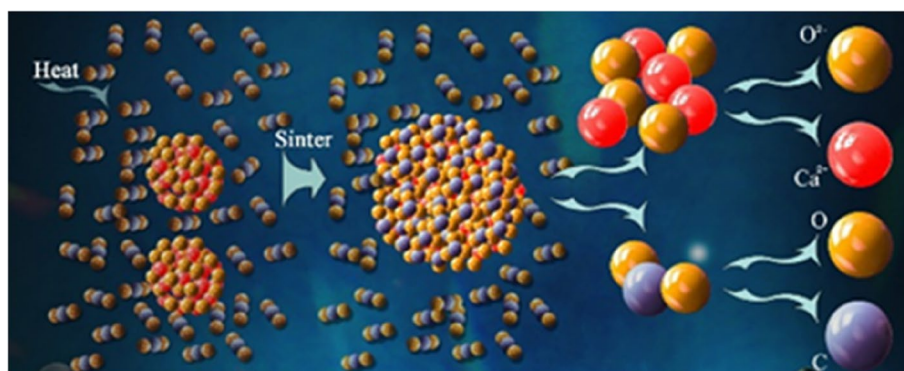


Fig. 10 The sintering mechanism of CaO materials during TCES process [213]

of modified materials at high carbonation pressure and long calcination/carbonation time. Al_2O_3 was mechanically milled as the dopant. Interestingly, the calcination/carbonation time was not fixed at long cycles. The CaO conversion was above 0.8 after 500 cycles, which is shown in Fig. 11.

High valence metallic oxides are also suitable dopants, such as SiO_2 , ZrO_2 and TiO_2 . Chen et al. [218] mechanically mixed limestone and nano- SiO_2 and the $X_{\text{ef},1}$ of modified material was 0.8. Its effective conversion dropped 22.5% after 20 cycles. Benitez-Guerrero et al. [219] prepared CaO/ SiO_2 composites by using rice husk as the biomass template. The $X_{\text{ef},1}$ was 0.48 and a 50% decrease of effective conversion occurred after 50 cycles. Wang et al. [220] used nano-calcium carbonate and tetrabutyl titanate as raw materials to obtain TiO_2 coated CaCO_3 material by the solution impregnation method. The CO_2 absorptivity of CaO decreased from 0.56 to 0.41 after 60 cycles. Xu et al. [221] also obtained CaCO_3 doped with TiO_2 by the solution impregnation method. The energy storage density decreased from 1257 kJ/kg to 798 kJ/kg after 30 cycles. Metal salts and non-metallic materials are also used as dopants to improve cyclic stability. Han et al. [222] synthesized high-performance calcium-based composites by adding expanded graphite to the CaO/ CaCO_3 energy storage system by the solution impregnation method. The energy storage density after 1st cycle was 1720 kJ/kg and it decreased to 1313 kJ/kg after 50 cycles, which was still a high value. Tian et al. [223, 224] investigated the modification mechanism of dopants in CaO/ CaCO_3 TCES system and synthesized Ti/Al/Mg co-doped calcium-based composites by a modified sol-gel method, where metal salts and metallic oxides work synergistically to improve the long-term cyclic stabilities. Their average energy storage densities exceed 1132 kJ/kg over 100 calcination/carbonation cycles and more than 75% of CaO still maintains reactivity after 100 cycles.

The cyclic stability of the CaO/ CaCO_3 TCES system has been widely investigated. Metallic oxides, metal salts and non-metallic materials are used as dopants to improve cyclic stability. These dopants prohibit sintering characteristics, maintain the porous structure, and improve the heat and mass transfer of calcium-based materials, which make calcium-based materials keep a high specific area and porosity. Then CaO can more easily react with CO_2 and the effective conversion is increased. However, the modification effect is limited and the effective conversion still decreases to certain content at long cycles. Therefore, it is necessary to reveal the modification mechanism at the atomic scale and raise the active design principle of high-performance materials.

Reaction kinetics According to the reversible property of chemical reactions, it is necessary for the initiation of the calcination reaction that the equilibrium pressure of CaCO_3 is greater than the CO_2 partial pressure in the system. The equilibrium pressure is calculated by the following expression [225]:

$$P_{\text{eq}} = \exp\left(17.74 - 0.00108T + 0.332\ln T - \frac{22020}{T}\right) (\text{atm}) \quad (7)$$

The equilibrium temperature of pure CO_2 at atmospheric pressure is about 895 °C [226]. Therefore, a high CO_2 concentration is desirable for increasing the heat release efficiency of the power-cycle because it allows carbonation at higher temperatures and faster rates. Except for the thermodynamic equilibrium, the kinetics of the reaction describes the rate of progression of the reaction, which is also an important parameter.

Reaction kinetics is also an important research direction of the CaO/ CaCO_3 TCES system. Incomplete calcination of CaCO_3 will lower the later effective conversion during the carbonation process. Therefore, it is necessary to make sure that decarbonation is complete. The addition

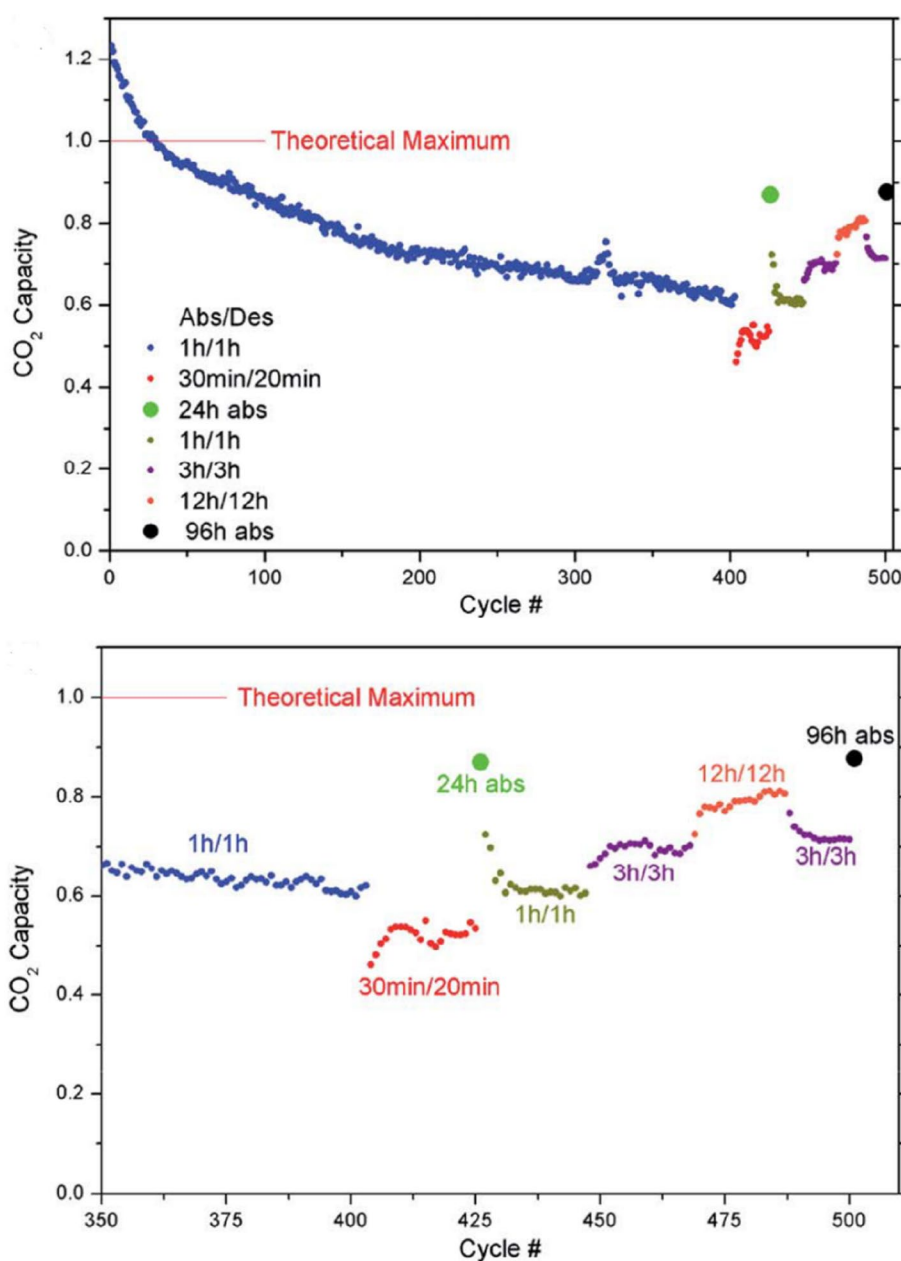


Fig. 11 The CO₂ capacity of Al₂O₃ doped CaCO₃ materials at long cycles [217]

of dopants can increase the decarbonation rate. In addition, CaO carbonation is a two-stage process: the first stage is the rapid surface reaction between solid and gas particles. The second stage is the diffusion-controlled reaction, which occurs when the product layer of CaCO₃ approaches a critical thickness. According to the gas-solid reaction model, the formation and growth of the CaCO₃ product layer is often considered to be the most important rate-limiting reaction step [227]. The layer of CaCO₃ impedes the ability of CO₂ to further diffuse to

the bulk and will also result in the plugging of the porous structure. The reactivity of the CaO and CO₂ particles can be improved by hydration of CaO, thermal pretreatment, and use of inert materials.

The kinetics of the CaCO₃ decarbonation have been widely investigated, but the activation energy and pre-exponential factor are not completely consistent in different studies [228]. Many researchers aimed to lower the activation energy to make CaCO₃ decomposition

easier to occur. Chen et al. [218] found that the specific heat capacity was improved by 20% due to the addition of SiO_2 . An optimal mass ratio of 5% SiO_2 showed a decrease in the activation energy by approximately 40 kJ/mol. Song et al. [229] found that doping Al and Fe elements into CaCO_3 powder reduced the decomposition temperature by 15 K to 24 K depending on the atmosphere. The activation energy had a sharp decrease when the decomposition reaction occurred in pure CO_2 atmosphere. Xu et al. [221] also found similar results. They decreased reaction activation energy of pure CaCO_3 from 1117.39 kJ/mol to 997.60 kJ/mol, and the onset decarbonation temperature from 903.56 °C to 876.13 °C in a CO_2 atmosphere by doping TiO_2 . Khosa et al. [230] decreased the decomposition temperature of CaCO_3 from 750 °C to 700 °C by doping SiO_2 . In addition, Khosa et al. [215] investigated the effect of doping ZnO. The lowest temperature at which the system can efficiently store energy was reduced by 25 °C from 750 to 725 °C after doping ZnO because of its high thermal conductivity. Arcenegui-Troya et al. [231] found that the presence of steam during calcination can reduce the apparent activation energy from 175 kJ/mol to 142 kJ/mol with a steam partial pressure of 29%. In general, reaction kinetics research is relevant to reaction barriers and many researchers aimed to decrease the decomposition activation energy to improve reaction rates.

Although the energy storage density of MgCO_3 is relatively high, it is difficult for actual application because of its low reaction kinetic characteristics. The interaction of MgO with CO_2 is kinetically prohibited which makes the heat release practically impossible [232]. A few researchers investigated the MgO/MgCO_3 TCES system to improve the carbonation rate. As shown in Fig. 12,

Shkatulov et al. [207] improved the energy storage performance of the $\text{MgO}-\text{CO}_2$ working pair by MgO modification with inorganic salts to promote the TCES dynamics. They found that CH_3COOLi and $\text{Li}_{0.42}\text{K}_{0.58}\text{NO}_3$ considerably promote the MgO carbonation. The material $\text{Li}_{0.42}\text{K}_{0.58}\text{NO}_3/\text{MgO}$ showed an increase in the carbonation conversion for the first 5 cycles and then had a stable de-carbonation conversion over the last 8 cycles. The heat storage capacity of the salt-promoted MgO was estimated to be 1.6 GJ/m³. The MgO/MgCO_3 working pair is of interest for chemical heat pumping. In addition, Xu et al. [233] studied the reaction temperature zone and exergy loss of the MgO/MgCO_3 TCES system. The exergy loss with dopant $\text{Na}_{0.5}\text{K}_{0.5}\text{NO}_3$ of 1: 0.1 can be optimized to 9.97 kJ/mol with an exergy efficiency of 91.0% while the original exergy loss was 44.75 kJ/mol with an exergy efficiency of 69.2%.

Reactor applications The energy storage of MgCO_3 is similar to $\text{Mg}(\text{OH})_2$, but only a few experimental studies focused on it. Its cyclic stability, reaction kinetics and heat and mass transfer still need to be further studied. Investigations on its chemical reactors have not been reported.

In recent years, it was widely reported to use CaO/CaCO_3 TCES for TCHS in CSP systems [234]. The system integration diagram of CSP coupled with CaL process is shown in Fig. 13 [209].

Although some CSP-CaL process schemes have been theoretically studied, the experimental studies of actual reactors have been less reported. Xu et al. [221] analyzed the performance of the CaCO_3/CaO TCES system in a fixed-bed reactor. They found the reaction temperature and

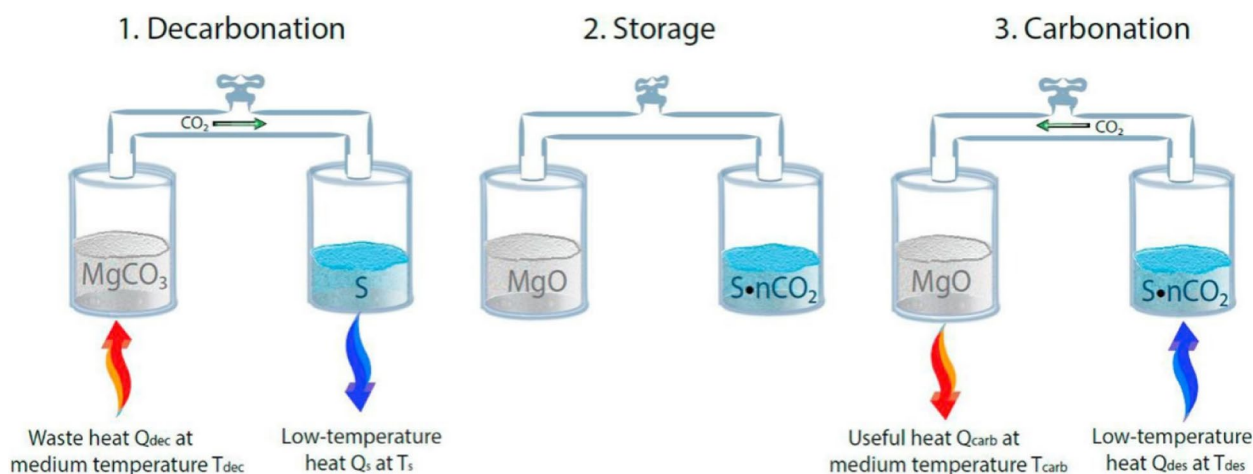


Fig. 12 An operation scheme of a chemical heat pump (heat storage mode) utilizing the $\text{MgO}-\text{CO}_2$ working pair [207]

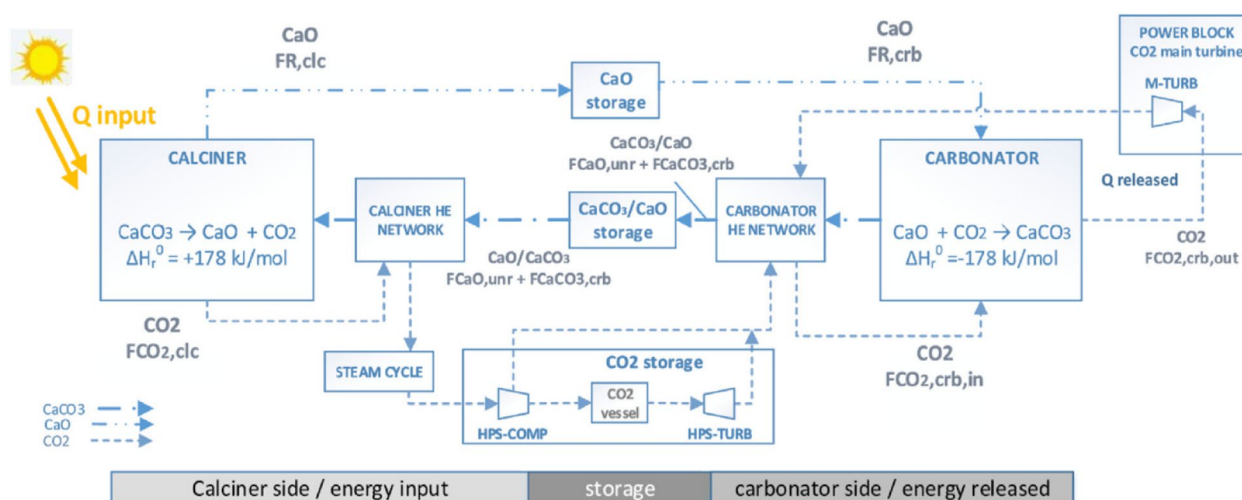


Fig. 13 CSP-CaL integration conceptual scheme and representation of energy storage and release processes [209]

reaction time limited the conversion rate of decarbonation. Chen et al. [235] designed a spiral coil reactor based on the combination of pneumatic transmission and secondary flow for CaO/CaCO₃ TCES. The CaO conversion was around 60–68%, which was a little higher than that in the fluidized bed reactor under the same conditions. Ma et al. [236] discussed the effects of the CO₂ concentration in carbonation, calcination/carbonation temperatures, and fluidization velocity in carbonation, particle size, and cycle number on the energy storage capacity and attrition behavior of limestone under fluidization. With increasing CO₂ concentration from 80 to 100%, the energy storage capacity and attrition rate of limestone increased by 11% and 9%, respectively. Lauro et al. [237] investigated the CaL-CSP integration in a lab-scale directly irradiated fluidized bed reactor. They improved the performance of the CaL-CSP system by lowering the calcination temperature, pre-calcination and using dolomite instead of limestone. Some fixed-bed and fluidized bed reactors of the CaO/CaCO₃ system have been studied, but the heat and mass transfer in the reactor still need to be further investigated because of the low conductivity of calcium-based materials [238]. The structural design and topological optimization of the reactor is a promising research direction.

Other carbonates proposed for TCS applications are mainly PbCO₃, BaCO₃ and SrCO₃ [234]. The decarbonation temperature of these carbonates are extremely high so these systems present low interest for further development compared with calcium-based materials. The calcination temperature of BaCO₃ reaches 1200–1400 °C and melting problems of the eutectic phase BaCO₃/BaO prohibits the carbonation reaction, which limits its

applicability for commercial TCES [157]. In addition, calcination/carbonation reactions of SrCO₃/SrO have been recently reported as a promising route for high temperature solar heat storage [239]. The calcination temperature is around 1200 °C, and presents a high energy storage density of 1585 kJ/kg. However, as in the case of CaO, SrO particles suffer sintering processes owing to the high working temperatures, which hampers the carbonation reaction due to a decrease of an active surface. On the other hand, although the PbCO₃/PbO system has been studied for chemical heat pump applications [158, 240], the characteristics of poor reversibility and toxicity of the products limit its application. Therefore, it has had few experimental studies.

4.2.4 Metal oxides

Redox reactions between metal oxides of different valence states can also be used for TCES. The first task in developing the TCS system based on the redox reaction is to find suitable metal oxides. Wong B used the HSC chemical reaction and equilibrium software to calculate the equilibrium thermodynamics of 74 different pure metal oxides, in which nominal air (79% N₂ and 21% O₂) at atmospheric pressure was used as the carrier gas [241]. This work screened some important redox systems, as shown in Table 12.

Considering the operating temperature (>700°C), materials costs, and preliminary experimental test data, the most studied materials were cobalt oxides [242], manganese oxides [243], barium oxides [244], copper oxides [245], iron oxides [246], and mixed metal oxides of these elements [247–249]. The basic heat storage properties of these pure metal oxides are shown in Table 13.

Table 12 Candidate materials for the TCS system based on the redox reaction [241]

Materials and Reaction	Temperature (°C)	Reaction Enthalpy (kJ/mol)	Energy Density (kJ/kg)
$2\text{PbO}_2 \rightarrow 2\text{PbO} + \text{O}_2$	405	62.8	262
$4\text{MnO}_2 \rightarrow 2\text{Mn}_2\text{O}_3 + \text{O}_2$	530	41.8	481
$2\text{BaO}_2 \rightarrow 2\text{BaO} + \text{O}_2$	885	72.5	474
$2\text{Co}_3\text{O}_4 \rightarrow 6\text{CoO} + \text{O}_2$	890	202.5	844
$6\text{Mn}_2\text{O}_3 \rightarrow 4\text{Mn}_3\text{O}_4 + \text{O}_2$	1000	31.9	202
$4\text{CuO} \rightarrow 2\text{Cu}_2\text{O} + \text{O}_2$	1120	64.5	811
$6\text{Fe}_2\text{O}_3 \rightarrow 4\text{Fe}_3\text{O}_4 + \text{O}_2$	1400	79.2	496
$2\text{Mn}_3\text{O}_4 \rightarrow 6\text{MnO} + \text{O}_2$	1700	194.6	850

In addition, perovskite, as a material with oxidizing and reducing properties, has also attracted the attention of researchers in the field of TCS [250–252].

Cyclic stability Cobalt oxides

Although cobalt oxide is expensive and toxic, due to its high theoretical energy storage density (844 kJ/kg) and heat release temperature (~900 °C), Co_3O_4 has been widely studied in the TCS system based on the redox reaction. For pure Co_3O_4 , it can be seen from Agrafiotis's [253] earlier experimental results (Fig. 14) that the pure Co_3O_4 powder has no obvious performance degradation after 30 cycles at 785 °C~985 °C, which meets the requirements of laboratory scale cycle stability. Wong [241] conducted 500 cycles of tests on Co_3O_4 at 700 °C~900 °C. SEM results showed that the decrease of cycle stability mainly occurred in the first 50 cycles, during which Co_3O_4 grains significantly grew and the grain size increased 5 times. They also found that mixing other metal elements is an effective way to improve the cyclic stability of cobalt oxide.

There are mainly two methods to improve the cyclic stability of cobalt oxide, namely morphological modification and chemical doping. The morphological modification

method is inspired by the application of porous metal foam in the field of heat exchange. The German Aerospace Center [254] designed a new heat storage material with a porous structure. It uses porous, redox-inert SHS material as the skeleton, and its outermost layer is covered by cobalt oxide. The author carried out 100 cycles of testing on 64wt% Co_3O_4 -loaded cordierite foam. After calculation, the conclusion was reached that the weight loss rate is stable at 6.64%, which corresponds to the weight loss rate after Co_3O_4 is fully reacted. It was further shown that redox properties are equal to those of the coated cobalt oxide at 100 cycles at 5 °C/min in the range of 800~1000 °C. For chemical doping, several studies have confirmed that Fe promotes the cyclic stability of $\text{Co}_3\text{O}_4/\text{CoO}$. Wong [241] tested the composite of $\text{Co}_3\text{O}_4 + 23.3\text{wt} \% \text{Fe}_2\text{O}_3$. At a heating/cooling rate of 30°C/min in the temperature range of 30–1000 °C, the thermal energy storage capacity of the 500th cycle was kept 60%. However, although a higher Fe_2O_3 content can improve the cycle stability and reduce the material cost, it also reduces the energy storage density. Andre [255] studied Co/Fe mixed metal oxides, and confirmed that a small amount of addition (5% Fe) is also beneficial to cycling properties.

In summary, $\text{Co}_3\text{O}_4/\text{CoO}$ is the most concerned metal oxide in TCES. Despite the shortcomings of high toxicity and high price, $\text{Co}_3\text{O}_4/\text{CoO}$ can withstand long-term heat storage and release cycles under different settings and different compositions or structures.

Manganese oxides

Due to the multivalent nature of Mn and the rich element content in the earth's crust, manganese-based metal oxides have also received attention from researchers. The conversion of Mn_2O_3 to Mn_3O_4 occurs at 950 °C, and the corresponding heat absorption is 202 kJ/kg Mn_2O_3 ; the conversion of Mn_3O_4 to MnO occurs at 1500 °C, and the corresponding heat absorption is 897 kJ/kg Mn_3O_4 . According to current survey results, the reversibility of

Table 13 Reaction temperature region, enthalpy and heat capacity of metal oxides

Temperature	Materials	Reaction Enthalpy (kJ/mol)	^a Heat capacity J/(mol·K)	Refs
700 °C < T < 900 °C	Co_3O_4	202.5	56.04~57.63	[242]
	BaO	72.5	56.93~58.54	[244]
900 °C < T < 1100 °C	Mn_2O_3	31.9	143.63~150.52	[243]
	CuO	64.5	56.75~58.39	[245]
1100 °C < T	Fe_2O_3	79.2	141.28(1100 °C)	[246]
	Mn_3O_4	194.6	212.37(1100 °C)	[243]

^a All heat capacity data correspond to the temperature in the first column

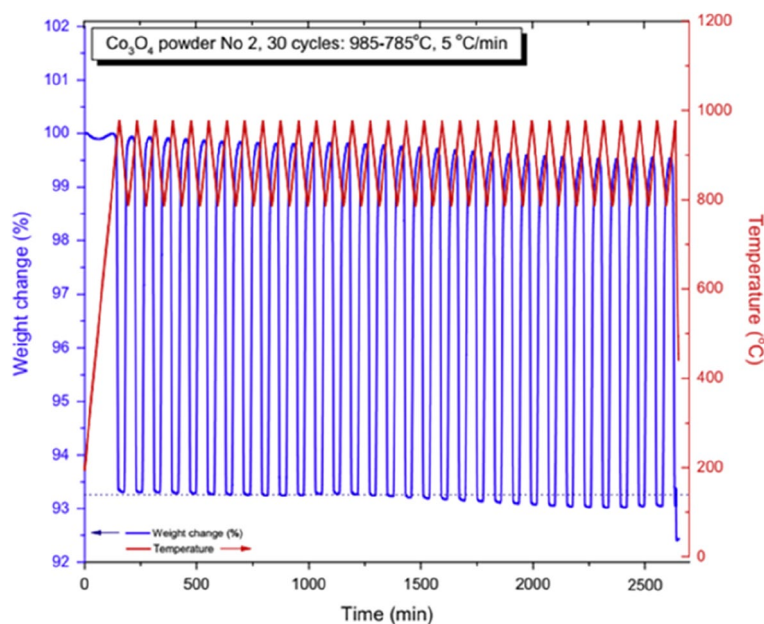


Fig. 14 30 cycles experiment of Co_3O_4 powder in the TGA between 985 and 785 °C with a heat/cooling rate of 5 °C/min. (Weight loss: 6.75% for the 1st cycle vs. 6.50% for the 30th cycle) [253]

the reaction of $\text{Mn}_3\text{O}_4/\text{MnO}$ has not been reported in the literatures [256]. Considering the high reaction temperature of $\text{Mn}_3\text{O}_4/\text{MnO}$, it requires high thermal insulation performance of the heat storage system. Therefore, it is difficult to use $\text{Mn}_3\text{O}_4/\text{MnO}$ for heat storage at this stage. In contrast, $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$ is a more suitable choice. However, $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$ has a low energy storage density, slow oxidation rate, unstable cycle performance, and 200 °C thermal hysteresis, all of which cause difficulties in its application in heat storage [257]. The current methods to solve the cycle stability of $\text{Mn}_2\text{O}_3 / \text{Mn}_3\text{O}_4$ are mainly chemical modification and morphological modification.

In terms of morphological modification, Carrillo [258] studied the effect of particle size on the cycle performance of $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$. It was found that the smaller the particle size, the larger the specific surface area, the higher the surface energy, and the more serious the sintering phenomenon. Therefore, reducing the particle size of $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$ is not beneficial to TCES. In order to improve the cycle stability, Agrafiotis [259] prepared foam Mn_2O_3 with reference to the method in reference [254]. However, the experimental results showed that this porous structure completely collapsed after high temperature sintering, and the material showed poor reversibility. The chemical modification of $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$ is mainly cationic doping. The doping of Co and Al proved to be of no benefit to the cycle stability

of $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$. Fe doping is the best way to improve the heat storage performance of $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$ [260]. Carrillo [261, 262] found that the doping of Fe can speed up the oxidation reaction rate. The cycle is more stable and thermal hysteresis reduce.

In summary, the energy storage density of manganese-based metal oxides is low, and the oxidation reaction rate is not high. However, its cycle stability can be improved by doping with metal cations, and lower cost is also an advantage of its use as a heat storage material. Finding the right dopant materials and proportions as well as the right morphology and structure is the key to applying $\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$ to heat storage technology.

Other oxides

Barium oxide was the first metal oxide redox couple to be experimentally studied for TCES performance [175]. Work done on the BaO_2/BaO redox couple is very limited, and all the experiments are carried out in TGA. In addition, only one literature has reported 200 cycles of barium oxide, and its conversion rate after long-term cycles is about 60% [244]. Alonso et al. [263] found that air is not conducive to reduction and particle sintering hinders the reaction. Optimization of reactor operation parameters may further improve the cycle performance. Similar results were reported by Deutsch et al. [245]. Fan

et al. [247] found that the addition of Mn_2O_3 powder into CuO can produce 30 consecutive charging or discharging operating conditions without any significant reduction in performance. The long-term stability of CuO has yet to be determined. Similar to copper metal oxides, iron has the advantage of being abundant in the earth's crust and lower in cost. Block [246] found that after three oxidation reductions, the oxide powders strongly sintered. Therefore, the possibility of implementing a heat storage system with iron-based oxide as the main material is low, but iron-based oxide can still be used as an additive to mix with other oxides for heat storage due to its low-cost advantage [264]. Al-Shankiti et al. [265] investigated the energy storage performance of Fe_2O_3 doped by 33% Mn_2O_3 . According to our survey result, there are few studies on the long-term cyclic stability of iron-based oxides. This has not been a research hotspot in recent years because the reaction temperature of iron-based oxides is too high to be put into practical application. However, with the increasing demand for high heat storage temperature, iron oxide will be a potential heat storage material in the future. Perovskite is a mixed metal oxide with an ABO_3 configuration, where A is usually alkaline earth or rare earth metal elements with a large ion radius, and B is usually transition metal elements with a smaller ionic radius [266]. With the development of high temperature resistant devices and the requirement of high efficiency of solar energy utilization, the temperature of centralized solar thermal power plant will be higher in the future. Perovskite has been proposed for TCES in recent years due to its diverse crystal structure, rich element composition, and wide range of temperature application [250, 252, 267]. The cycling performance of perovskite is obviously superior to that of iron and copper based oxides. However, in perovskite systems,

the reported energy density is usually less than 300 kJ/kg, making it less competitive in this respect than other REDOX systems. There is room for further improvement in the cyclic stability of these oxide.

To sum up, there are two research ideas for improving cyclic stability: 1) mixing metal elements, and 2) using reactors with efficient heat and mass transfer. The first idea starts from the material itself and is the current research hotspot.

From the perspective of the phenomenon, the doping of an appropriate proportion of the second or third (or even fourth) metal elements has a certain promotion effect on the cycling performance, meanwhile the doping also causes the reaction temperature, energy storage density, and reaction rate at the level of the pure metal oxide to deviate. From the perspective of mechanism, most of the current researches only explain the modification results based on microscopic characterization data, but the understanding of the active law of different doping elements is absent. In order to realize the active design of a heat storage metal oxide with stable cyclability, medium to high energy storage density, suitable operating temperature, and fast reaction rates, it is necessary to conduct in-depth research on the mechanism of material modification. The study of internal laws is very important by means of first-principal calculations at atomic scale in the future.

Reaction kinetics Table 14 lists some recent kinetic studies of the redox couples. In most metal oxide heat storage systems, the oxidation rate is slower than the reduction rate, which is also one of the main reasons for the poor long-term cycle stability of the material. Compared with the pure oxide REDOX system, the kinetic

Table 14 Recent studies on reaction kinetics of metal oxide redox reactions

Year	Oxides	Kinetics ^a	Refs
2015	Co_3O_4	Reduction (Avrami–Erofeyev nucleation model): an Avrami constant of 1.968 and apparent activation energy of 247.21 kJ/mol at 1113 K–1213 K Oxidation: apparent activation energy of 58.07 kJ/mol at 450 K–750 K	[268]
2019	MnFe_2O_4	First by a diffusion-controlled reaction mechanism ($E_a = 192 \pm 2 \text{ kJ/mol}$) with no phase change, followed by a nucleation-growth reaction mechanism ($E_a = 181.4 \pm 0.3 \text{ kJ/mol}$)	[265]
2019	Mn + Fe oxide	For reaction in Ar between conversions $0.05 < \alpha < 0.95$ and reaction in air in the conversion range of $0.05 < \alpha < 0.8$: $\frac{d\alpha}{dt} = \frac{(8.85 \pm 0.00)10^{15}}{\beta} \exp\left(-\frac{426.13 \pm 0.04}{RT}\right) \frac{3(1-\alpha)^{2/3}}{2[1-(1-\alpha)^{1/3}]} \left(1 - \frac{P_{\text{O}_2}}{P_{\text{O}_2,eq}}\right)^{25.01}$	[260]
2021	Si doped Mn_2O_3	Reduction: $\frac{d\alpha_{\text{red}}}{dt} = 3.8 \cdot 10^{15} \cdot e^{\left(-\frac{341.72}{RT}\right)} \alpha^{0.1} \cdot (1 - \alpha_{\text{red}})^{1.15} \cdot [-\ln(1 - \alpha_{\text{red}})]^{0.05} \cdot \left(1 - \frac{P_{\text{O}_2}}{8 \cdot 10^7 \cdot e^{-23.13} \left(\frac{1000}{T}\right)}\right)^{7.05}$ (Sestak-Berggren reaction model) Oxidation: $\frac{d\alpha_{\text{ox}}}{dt} = 1.7 \cdot 10^7 \cdot e^{\left(-\frac{-120.42 + 156.88 \frac{P_{\text{O}_2,eq}}{P_{\text{O}_2,eq} - P_{\text{O}_2}}}{RT}\right)} \cdot \alpha^{2.7} \cdot (1 - \alpha_{\text{ox}})^{0.5} \cdot [-\ln(1 - \alpha_{\text{ox}})]^{-1.85}$	[269]

^a $\frac{d\alpha}{dt} = k(T) \times f(\alpha) \times h(P_{\text{O}_2})$, $k(T) = Ae^{-\frac{E_a}{RT}}$, where $\alpha(t)$ is the mass change with time, E_a is the activation energy, A is the pre-exponential factor, $f(\alpha)$ is the reaction model, and $h(P_{\text{O}_2})$ is a pressure-related term

study of the mixed metal oxide REDOX system is the current research focus, and only a few studies have done a complete kinetic analysis of the REDOX reaction of perovskite in the field of heat storage. In addition, similar to the research on improving the cycle stability of materials, most of the researchers doped metal elements or mixed metal oxides together to increase the oxidation rate. However, the selection of dopant materials seems to be random and lacks the explanation of its internal mechanism and the summary of the law.

Reactor and application The research on TCES based on the redox reaction is still in the stage of laboratory materials research, and there are few studies on pilot-scale reactors. However, with the further development of material cycling and stability research, the detailed design and development of reactors will receive more and more attention. At present, most TCES studies are mainly combined with solar thermal power generation and photovoltaic power generation systems. The aim is to solve the problem of intermittent supply due to environmental factors.

Tescari et al. [270] designed a reactor with a honeycomb metal oxide. High-temperature gases pass through the honeycomb and insulation is used outside the reactor to reduce heat loss. The honeycomb structure is conducive to gas flow and a more uniform temperature distribution. However, the long-term cyclic stability of honeycomb structures is not easy to maintain. The design has a limited heat and mass transfer rate. Wokon et al. [257] designed a reactor housed in a tubular furnace as an auxiliary heat source to minimize heat loss. It has the advantages of easy operation, a wide temperature range, and

lower attrition. However, the problems of poor heat and mass transfer, uneven distribution of temperature along the axis, and slow reaction rate still exist. Hamidi et al. [271, 272] used experimental and numerical methods to calculate the effective thermal conductivity of the reactor. Metal oxide particles are packed in the reactor chamber; heaters are placed on the side of the reactor; and gas flows from top to bottom, as shown in Fig. 15. Wu et al. [264] designed an air-fed fluidized bed integrated with a solar receiver shown in Fig. 16. Materials are dropped in the solar receiver and then oxidized in the fluidized bed. Released heat is used for driving a supercritical CO₂ cycle. The fluidized bed may reduce the reversibility loss of metal oxide. However, it has a medium pressure drop and the system is complex. Alonso et al. [263] designed a moving bed reactor. The reaction chamber keeps rotating and enhances the mixing of reactant solids. It is easy to obtain a homogeneous temperature. The concentrator helps reduce radiation heat loss, and it has good heat and mass transfer preventing sintering. However, its energy loss is high because much energy is used to heat up the reactor rather than the reactant particles.

4.3 Summary

TCES can convert thermal energy into chemical energy. The heat can be stored at the ambient temperature for a long period due to thermal energy stored as chemical energy. Usually, the chemical reaction energy is larger than sensible heat and latent heat. TCES has the greatest energy density among the three thermal storage technologies, but it has a complicated control process in the application of heat storage. Furthermore, how to obtain a

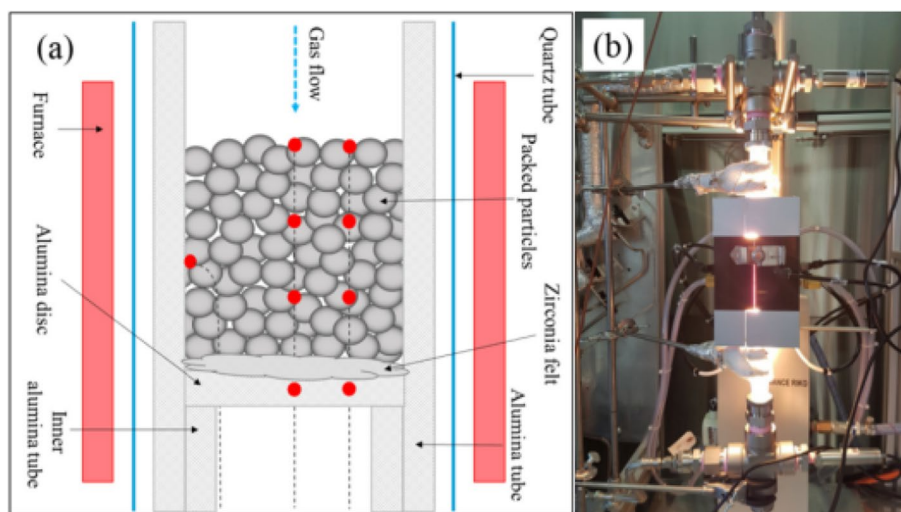


Fig. 15 The packed particles and the reactor chamber [271, 272]

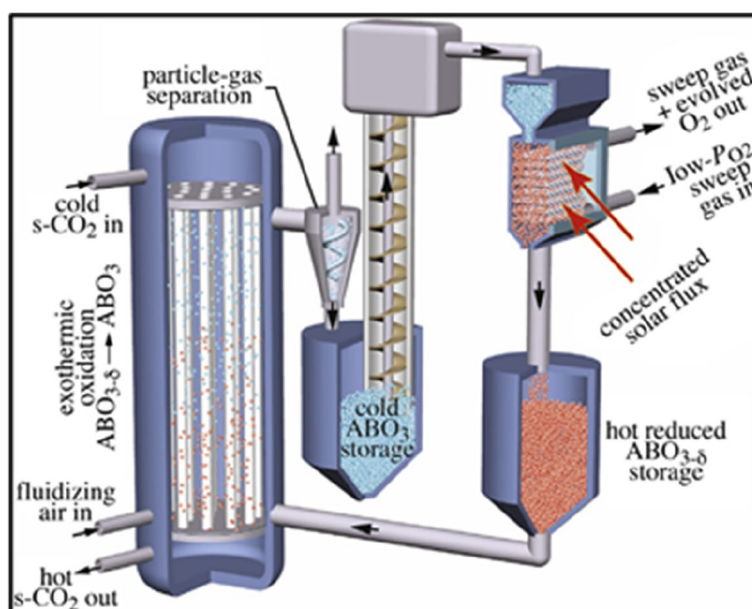


Fig. 16 The air-fed fluidized bed integrated with a solar receiver [264]

perfect material, which includes cyclic stability, low cost, and safety, is a big problem.

Metal hydrides, metal hydroxides, metallic carbonates and metal oxides were widely investigated and have good application prospects. Although metal hydrides have advantages of high energy density and wide temperature range, their operating condition is hard to be satisfied. In addition, the separation and storage of H_2 is also an important problem, which poses limitations to the development of this technology. Metal hydrides, especially $Ca(OH)_2$ and $Mg(OH)_2$ have advantages of low material cost, nontoxicity, and high intrinsic stability. Many researchers have studied their reaction kinetics, cyclic stability and heat and mass transfer in reactors. The main limitations are the low thermal conductivity, heat and mass transfer, and ease of being polluted by CO_2 . Therefore, the optimization of systems urgently needs to be studied. Among metallic carbonates, $CaCO_3$ has the advantages of high energy storage density, high reaction temperature, and low cost of materials. It is necessary to reveal the modification mechanism, and raise the active design principle of high-performance materials to achieve high cyclic stability. In addition, structural design and topological optimization of the reactor is a promising research orientation to improve heat and mass transfer. For metal oxides, material modification has a certain promotion effect on the cycling performance but it will cause the reaction temperature, energy storage density, reaction rate at the level of pure metal oxide deviation. In

addition, there are few studies on pilot-scale reactors of metal oxides.

5 Comparison and analysis

5.1 Comparison and analysis

As we reviewed above, there are lots of heat storage choices. Meanwhile, the requirements of heat storage properties (temperature range, heat storage capacity, materials cost, maturity, etc.) is various under different applications. Therefore, a comparison of some typical heat storage choices is listed in Table 15. The materials among three typical heat storage methods have been presented in the above analysis. Temperature shown in the table means the operating temperature range for the SHS materials, while the melting temperature for LHS and charging reaction temperature for TCES, respectively. Charging reaction temperature is the decomposition temperature of TCES materials under the condition that the partial pressure of the gaseous reactant is low. c_p represents the specific heat capacity. Δh is the enthalpy of phase change for LHS and reaction enthalpy for TCES, respectively. The calculation of the materials cost is based on market research in China (based the year 2022) and the above review (around the year 2010–2019 according to the references). Therefore, the materials cost is compared in RMB per kilowatt-hours ($¥/kW \cdot h$). The maturity of the heat storage methods is divided into three levels: commercial application (CA), pilot test (PT), and laboratory research (LR).

Table 15 Comparison of some typical heat storage methods

	Materials	T °C	c_p (kJ/kg·K)	Δh (kJ/kg)	Cost (¥/kW·h)	Maturity	Refs
SHS	Water	0–100	4.183	-	0.07	CA	[48]
	Mineral oil	200–300	2.6	-	26.8		[19]
	Synthetic oil	250–350	2.3	-	274		[19]
	Reinforced concrete	200–400	0.85	-	6.4		[19]
	Solar salt	220–565	1.5	-	37		[49]
	Carbonate salts	450–850	1.8	-	70		[19]
LHS	n-Hexadecane	18	-	237	577	CA; PT	[54–56]
	Acetic acid	16.7	-	184	156		[54–56]
	Lauric acid	44	-	218	248		[54–56]
	Catechol	104.3	-	207	1200		[54–56]
	$\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$	27.45–30	-	161.15–192	40–47		[56, 64]
	$\text{Na}(\text{CH}_3\text{COO}) \cdot 3\text{H}_2\text{O}$	58–58.4	-	226–264	72–85		[54, 66]
	$\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$	78	-	265.7–301	72–81		[65]
	$\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$	115–117	-	195–172	31–37		[65]
	88% Al–12% Si	576	-	560	193		[66, 73]
	LiNO_3	250–254	-	360–373	1880–1950		[56, 73]
	NaNO_3	306–310	-	172–199	90–105		[63]
	NaCl	800–802	-	466.7–492	6.2–6.5		[63, 73]
	NaCO_3	854–858	-	165–275.7	52–87		[73, 77]
TCES	Mg/MgH_2	279	-	2811 / (MgH_2)	90	PT; LA	[164]
	$\text{CaO/Ca}(\text{OH})_2$	440	-	1409 / ($\text{Ca}(\text{OH})_2$)	1.9		[200]
	$\text{MgO/Mg}(\text{OH})_2$	325	-	1389 / ($\text{Mg}(\text{OH})_2$)	4.5		[171]
	CaO/CaCO_3	890	-	1778 / (CaCO_3)	1.2		[210]
	MgO/MgCO_3	397	-	1381 / (MgCO_3)	3		[207]
	$\text{CoO/Co}_3\text{O}_4$	905	-	841 / (Co_3O_4)	2890		[241, 242]
	$\text{Mn}_2\text{O}_3/\text{Mn}_3\text{O}_4$	928	-	265 / (Mn_2O_3)	85		[241, 243]
	$\text{Cu}_2\text{O/CuO}$	1025	-	811 / (CuO)	277		[241, 245]

SHS has already realized commercial applications, especially for water and solar salt. Water is used widely in our daily life. Water is a good SHS material because of the great specific heat capacity and the low cost. However, the applicable temperature range is limited. Solar salt is often used in solar thermal power plants, but the corrosion of solar salt is a big problem. Except for high-temperature application in CSP, most applications of SHS are focused on the low-temperature range.

As shown in Table 15, LHS usually can offer high heat storage density compared to SHS, and the heat storage/release temperature of LHS can be maintained at a constant temperature. Therefore, the LHS has attracted more and more attention in both research and application field. Paraffin wax is widely used in lots of research works of low-temperature range because of the easy selection of required melting temperature and stable property. Most PCMs are still in the pilot test, especially for the high-temperature application. However, there are many commercial applications in medium- and low-temperature, especially for heating and domestic hot water.

$\text{Na}(\text{CH}_3\text{COO}) \cdot 3\text{H}_2\text{O}$ and $\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$ are often used by many Chinese companies because of the appropriate temperature and low cost, such as Pioneer Energy (Jiangsu) Co., Ltd, Jinhe Energy Co., Ltd, and so on. Though $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ and NaCl have lower cost, they are not widely used because of the strong corrosion of Cl^- . Some materials are very expensive which limit their application, such as LiNO_3 , catechol and so on.

TCES has a huge energy density which is usually above 1000 kJ/kg except for most redox reactions. In general, the complicated heat storage process results in the difficult commercial application of TCES. Besides the above problem, every material has itself problems. Except for the commercial application of chemical adsorption in the low temperature, most TCES technologies are still in laboratory research, and a few materials ($\text{CaO/Ca}(\text{OH})_2$, CaO/CaCO_3 , etc.) are in the pilot test. Redox reactions usually can be used in very high temperature application field, but the low energy density and high cost affect the commercial application. As shown in the Table 15, the Mg/MgH_2 has the maximal energy density 2811 kJ/kg

which is based on MgH_2 . However, the dangerousness of H_2 limits its application. $\text{CaO}/\text{Ca}(\text{OH})_2$, $\text{MgO}/\text{Mg}(\text{OH})_2$, CaO/CaCO_3 , and MgO/MgCO_3 are very promising materials for large-scale heat storage application because of the low cost, high energy density, and so on.

In a word, the SHS is the most mature technology, but the energy density is the lowest, which needs a very large space cost. The TCES has the largest energy density, which can save financial and space cost. Though the TCES technology is unmatured, the advantage of TCES is obviously, which can be realizing the large-scale heat storage application in the future. The maturity of LES technology is intermediate between SHS and TCES. Meanwhile, more LES technologies will be commercialized.

5.2 New application—Carnot battery

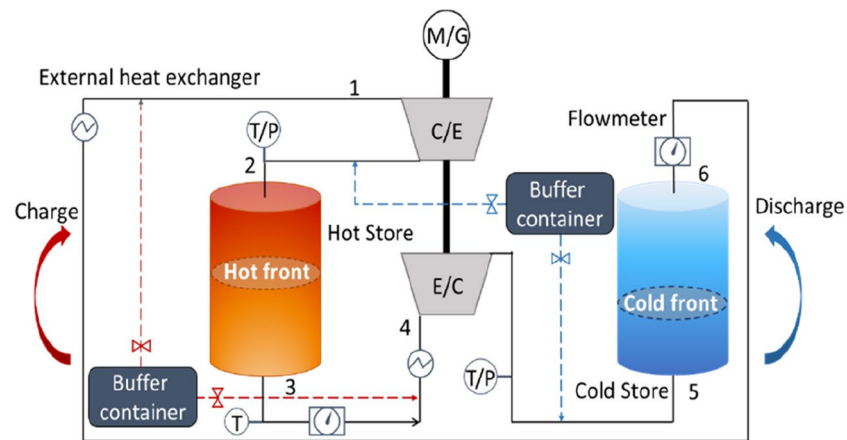
The Carnot battery is a new type of electric energy storage technology developed based on the thermodynamic cycle and thermal energy storage technology, also known as pumped-thermal electricity storage (PTES), which can make up for the shortcomings and deficiencies of traditional large-scale energy storage technology. The Carnot battery with waste heat recovery can greatly improve the overall roundtrip efficiency. It is a new type of energy storage/supply technology that can realize the “cold, heating, and electricity” co-supply. According to the cycle type, it can be mainly divided into three categories: Joule-Brayton cycle-based PTES, Rankine cycle-based PTES, and Transcritical cycle-based PTES, which each have their own application scenarios.

The Joule-Brayton cycle-based PTES has a simple operating principle and can be applied to high-temperature thermal Energy Storage. During the charging period, the electric motor is driven by excess electricity supplied by the grid or generated by renewable energy; the motor compresses the working fluid to high-temperature and high-pressure. The working fluid flows through the heat storage tanks, storing heat in the sensible storage materials or PCMs. Then, it is expanded to ambient pressure and low-temperature, storing the cold thermal energy in the cold tanks. In peak demand hours, the working fluid flows in the opposite direction, absorbing the heat/cold energy stored in tanks to drive the generator to generate electricity. The working fluids are mainly air, helium, and argon. Ge et al. [273] established a 10.5 MW/5 h energy storage model and found that the energy density of latent storage increased from 232.5 kWh/m³ to 245.4 kWh/m³ compared to sensible storage, as shown in Fig. 17a. Xue et al. [274] optimized the design of the system by replacing the external heat exchanger between the hot store and the expander with a recuperator as shown in Fig. 17b, and found that the energy density of the system was further

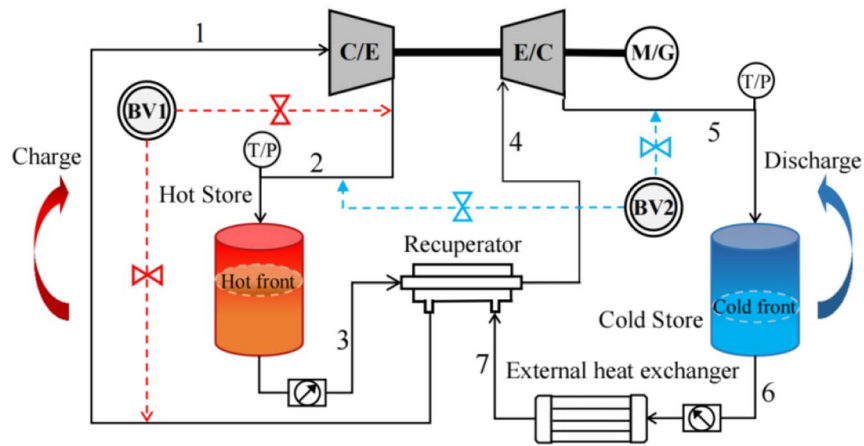
increased to 267.4 kWh/m³. In addition, they also analyzed the dimensionless temperature distribution in the packed-bed latent heat/cold stores and the exergy flow of the system and concluded that the exergy efficiency of this recuperated Carnot battery could reach 55.3%. Zhao et al. [275] further divided exergy loss into avoidable, unavoidable, endogenous as well as exogenous destruction, and found that the avoidable exergy loss of cold heat exchanger is the greatest. Wang et al. [276] studied the delivery stability of Brayton cycle-based Carnot battery operated in four modes, including series, parallel, in-sequence, and the “temperature complementation”. They found that delivery stability of the system was further improved in the fourth mode, with a variation of 13.2%. A complete and efficient Joule-Brayton cycle-based PTES laboratory has not yet been established. The Newcastle team [277] tried to assemble and commission an energy storage system with a rated power of 150 kW and a storage capacity of 600 kWh at the Sir Joseph Swan Energy Research Center, as shown in Fig. 17c.

The Rankine cycle-based PTES has high energy storage density, desired long life span (>20 years), and mature technology in terms of thermodynamics as well as equipment manufacturing. In the case of integrating with industrial waste heat, its round-trip efficiency can exceed 1. This type of Carnot battery is widely used in low- and medium-temperature situations, especially in the application of waste heat recovery. Hassan et al. [278] conducted a numerical analysis of the Rankine cycle-based PTES system, and the roundtrip efficiency of this system can reach 1.34 under the heat source and organic Rankine cycle (ORC) sink temperatures of 100 and 25 °C, respectively. Xue et al. [279] performed a multi-objective analysis of organic Rankine cycle-based Carnot battery and found that the roundtrip efficiency and exergy efficiency of the system could reach 0.97 and 0.64 at the same time. The application of this type of Carnot battery in the coal-fired power plants was also explored and the results show that when the rated power of the power plant reaches 200 MW, the power peak shaving capacity of the coal-fired power plant integrated with Carnot battery can reach 94.4%. Steinmann et al. [280] explored the role of Carnot batteries in a smart district energy system for seasonal heating and power management. DLR (Stuttgart, Germany) [281] prepared a laboratory for the Rankine cycle-based PTES system. A Temperature-controlled heat source and radiator and the electrical connections of various components were installed. They will connect the high-temperature heat pump (HTHP) prototype and the ORC prototype in the next stage of work.

The Transcritical cycle-based PTES uses CO_2 as the working fluid due to its good thermal performance,



(a)



(b)



(c)

Fig. 17 Representative Carnot battery system design and experimental layout: **a** PTES based on packed-bed latent heat/cold stores, **b** Brayton-like recuperated PTES system and **c** the PTES system at Newcastle's Joseph Swan Centre, with the two tanks filled with mineral particulate [273, 274, 277]

non-toxicity and low critical pressure characteristics. The acceptable roundtrip efficiency and relative low cost make it a potential large-scale energy storage system.

Although the heat storage part used in current researches of the Carnot battery system is SHS materials and PCMs, TCES can also be applied in the Carnot battery system. Zamengo et al. [282] investigated a Carnot battery based on chemical heat storage/pump (CHS/P) and concluded that CHS/P systems have a higher energy storage density and total storage volume is 66% smaller than that of SHS system. Saghafifar et al. [283] also studied a chemical looping electricity storage (CLES) system and found that the energy density of Carnot battery based on this type of thermal energy technology could have an energy storage density of 250–350 kWh/m³ and that it could achieve a roundtrip efficiency of 40–55%.

6 Conclusions

The influence of carbon dioxide (CO₂) on global warming has been demonstrated. China is committed to the targets of achieving peak CO₂ emissions around 2030 and realizing carbon neutrality around 2060. To realize carbon neutrality, human is trying to use renewable energy to take the place of fossil fuels. However, renewable energy, such as solar and wind power, has inherent intermittency and fluctuation properties. To settle that problems and improve the stability of renewable energy, the energy storage method is necessary. Considering that the final energy consumption of humans is most used in the form of thermal energy, thermal energy storage can have a good promising development in the future, and will be an important role in low carbon emissions. Therefore, the state-of-the-art researches in sensible, latent and TCES media were reviewed comprehensively from the renewable energy and the carbon neutrality perspective.

- (1) SHS is the most matured technology, but the low energy density usually results in a huge space cost. The great volume of the system will lead to bigger heat loss and more heat transfer areas. Furthermore, the corrosion of molten salt is another problem. The new materials and the novel heat transfer methods are the research highlights for SHS in the future.
- (2) A lot of materials can be used as PCMs, such as organic compounds (paraffin wax, fatty acids, etc.), inorganic salts and their eutectics (hydrated salt, metals and alloys, etc.) and so on. In addition, corrosion, cycling, and thermal stability issues of these inorganic salts and composite PCMs still need further study. Generally, PCMs have low thermal conductivity, which is a great challenge for the

application of LHS. Most PCMs are still in the pilot test, especially for the high temperature application. However, there are many commercial applications in medium and low temperature, especially for heating and domestic hot water.

- (3) TCES has the greatest energy density among the three thermal storage technologies, but it has a complicated control process in the application of heat storage. The main limitations of TCES materials are the low thermal conductivity, poor heat and mass transfer property, bad thermal cycling stability. The doping of an appropriate proportion of the second or third (or even fourth) metal elements has a certain promotion effect on the cycling performance, reaction rate, and so on. However, the selection of dopant materials seems to be random and lacks an explanation of its internal mechanism and the summary of the law. Though the TCES technology is unmaturing, the advantage of TCES is obviously, which can be realizing the large-scale heat storage application in the future.
- (4) The Carnot battery is a new type of electric energy storage technology developed based on the thermodynamic cycle and thermal energy storage technology. Although the thermal energy storage used in current researches of the Carnot battery system is SHS materials and PCMs, TCES can also be applied in the Carnot battery system.

Abbreviations

IPCC	Intergovernmental Panel on Climate Change
IEA	International Energy Agency
SHS	Sensible Heat Storage
LHS	Latent Heat Storage
PCMs	Phase Change Materials
TCES	Thermochemical Energy Storage
CSP	Concentrated Solar Power
SWCNTs	Single-walled carbon nanotubes
CLHS	Cascaded latent heat storage
MOFs	Metal Organic Frameworks
MH	Metal Hydrides
CA	Commercial Application
PT	Pilot Test
LR	Laboratory Research
PTES	Pumped-Thermal Electricity Storage
HTHP	High-Temperature Heat Pump
ORC	Organic Rankine Cycle

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Authors' contributions

C.Y. Zhao conceived of the study and edit the manuscript. J. Yan performed the data analyses and draft the manuscript. X.K. Tian, X.J. Xue, and Y. Zhao participated in the design of this work and played equally important roles in analyzing and summarizing the results. We ensure that all authors are included in the author list, and its order has been agreed by all authors. The author(s) read and approved the final manuscript.

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Availability of data and materials

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

Changying Zhao is the editor-in-chief of Carbon Neutrality and was not involved in the editorial review, or the decision to publish this article. All authors declare that there are no competing interests.

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